Secure Microfinance and Welfare Management Platform Using Blockchain and AI

INDIVIDUAL THESIS

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Individual component: Machine Learning-Based Loan Affordability and Repayment Risk Prediction Framework for Microfinance Institutions

Group ID: R26-IT-017

Report submitted in partial fulfilment of the requirements for the

Bachelor of Science (Honors) in Information Technology

Specialization in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

Sri Lanka

October 2026

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ABSTRACT

Access to financial assistance and microloans is essential for helping low-income communities improve their economic stability and livelihoods. However, many microfinance institutions and welfare programs struggle to accurately evaluate the repayment ability of loan applicants. Traditional applicant assessment methods rely heavily on manual reviews and limited, structured financial data, which often results in inaccurate risk evaluations, higher default rates, and lengthy loan approval delays. These shortcomings emphasize the critical need for an intelligent, data-driven decision-making framework. While several current financial systems offer basic loan management features, they lack the predictive capabilities required to analyze complex customer characteristics and estimate the probability of successful repayment.

To address this gap, this study introduces a Machine Learning-Based Loan Affordability and Repayment Risk Prediction Framework. The proposed solution analyzes a diverse set of borrower attributes, including applicant income, expenses, assets, employment status, share amount, requested loan amount, savings account balance, existing loans, and repayment history—to generate a repayment probability score and a standardized risk classification. Utilizing an anonymized, combined dataset of alternative credit scoring and microfinance loan records, the study evaluated multiple supervised classifiers, including Logistic Regression, Decision Trees, and Random Forest. Random Forest was selected as the primary predictive model due to its superior ability to capture non-linear relationships among financial attributes and reduce overfitting, ultimately achieving an accuracy of 80.5%.

Furthermore, the predictive model is paired with a rule-based decision recommendation module that computes safe monthly installments and suggests feasible loan types and repayment durations based on the borrower's calculated disposable income. Implemented using Python and data processing libraries such as Pandas and Scikit-learn, this integrated system provides real-time risk predictions during the loan application process. The expected result is a dependable, automated decision-support system that enhances the speed, accuracy, and fairness of loan approvals while significantly reducing institutional financial risk. Ultimately, this framework offers substantial social and commercial advantages by promoting responsible micro-lending practices and improving financial access for underserved rural communities

Keywords: Machine Learning, Loan Prediction, Repayment Risk Analysis, Microfinance, Random Forest, Decision Support System, Affordability Assessment

ACKNOWLEDGEMENTS

I would like to express my deepest gratitude to my principal supervisor, Dr. Dharshana Kasthurirathne, and my co-supervisor, Ms. Chathurya Kumarapperuma, for their continuous guidance, insightful feedback, and unwavering support throughout the development of this research project. Their expertise and encouragement were invaluable in shaping the direction of this study.

I also extend my sincere appreciation to the external stakeholders who provided essential resources for this research. Special thanks to Mr. Rangith Disanayaka, Director of the Co-operative Society Battaramulla, as well as the Manager of Samupakara Gramiya Bank Battaramulla and the representatives from Samurdhi Bank. The access they provided to real-world loan datasets, welfare-related data, and practical requirement specifications was critical for the successful training, development, and evaluation of the machine learning predictive models and the overall microfinance platform.

My sincere thanks also go to the academic and administrative staff of the Department of Information Technology at the Sri Lanka Institute of Information Technology (SLIIT) for providing the necessary academic foundation, resources, and environment required to conduct this research successfully.

Finally, I would like to thank my project group members for their collaboration and technical support during the integration phases, as well as my family and friends for their endless patience, encouragement, and support during the completion of this dissertation.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
ML	Machine Learning
AI	Artificial Intelligence
RF	Random Forest
SPSS	Statistical Package for the Social Sciences
DSS	Decision Support System
UI	User Interface
DB	Database
F1 Score	Harmonic Mean of Precision and Recall
SMOTE	Synthetic Minority Over-sampling Technique
ROC	Receiver Operating Characteristic
AUC	Area Under the Curve
MFI	Microfinance Institution
NPL	Non-Performing Loan
CBSL	Central Bank of Sri Lanka
API	Application Programming Interface
CART	Classification and Regression Tree
D-App	Decentralized Application

1 INTRODUCTION

1.1 Overall project context

Access to microcredit and structured financial assistance serves as a fundamental catalyst for socio-economic mobility, poverty alleviation, and community empowerment across developing nations. In emerging economies such as Sri Lanka, microfinance institutions (MFIs), cooperative rural development banks (such as Samupakara Gramiya Sanwardhana Banks) , and government-backed welfare schemes (such as the Samurdhi program, ASWASUMA) provide essential credit facilities to low-income earners, rural entrepreneurs, agriculturalists, and daily-wage laborers who are traditionally excluded from commercial banking ecosystems. Historically, commercial financial institutions evaluate creditworthiness using centralized institutional credit offices such as the Credit Information Bureau of Sri Lanka (CRIB) which track formal credit card histories, commercial debts, asset mortgages, and audited personal taxation filings. However, the vast majority of microfinance applicants operate within informal, cash-driven micro-economies where formal credit bureau footprints are non-existent.

Despite their societal importance, conventional microfinance operations remain severely hindered by manual workflows, fragmented paper ledgers, decentralized record-keeping, and vulnerability to fraudulent claims or duplicate identity submissions. To combat these systemic vulnerabilities, the overarching research initiative titled *"Secure Micro Finance and Welfare Management Platform Using Blockchain"* was conceived. This multidisciplinary project seeks to engineer a secure, decentralized, and transparent digital infrastructure designed to modernize community microfinance and welfare disbursement. The platform architecture integrates four synergistic technological pillars:

1. An immutable, distributed ledger framework utilizing blockchain technology to cryptographically secure loan issuance, repayment ledgers, and welfare transaction histories against unauthorized tampering.
2. A multilingual conversational artificial intelligence agent utilizing Retrieval-Augmented Generation (RAG) to provide natural-language accessibility in Sinhala, Tamil, and English regarding welfare eligibility, loan policies, and institutional procedures.
3. An automated identity verification and fraud-prevention module to eliminate ghost beneficiaries and duplicate loan enrollments.
4. An intelligent, automated credit risk evaluation and financial decision-support engine capable of assessing repayment viability in the absence of conventional credit bureau scores.

1.2 Individual component context

Within the ecosystem of the secure microfinance platform, this individual research component is titled:

"Machine Learning-Based Loan Affordability and Repayment Risk Prediction Framework for Microfinance Institutions"

While distributed ledgers provide cryptographic security and auditability for historical transactions, they do not inherently evaluate whether an applicant possesses the financial capacity to service a newly requested loan. A microfinance bank cannot rely solely on transaction logging; it requires forward-looking predictive intelligence at the point of application ingestion to safeguard its liquidity.

This component functions as the primary analytical and decision-making engine of the platform. When a borrower submits a loan request whether for agricultural expansion, house repair, micro-enterprise capitalization, or emergency consumption the framework ingests multidimensional financial and non-traditional behavioural attributes. These metrics include regular monthly earnings, secondary or gig-economy income, household living expenses, accumulated micro-savings balances, outstanding debt commitments, historical micro-loan repayment performance, and the availability of mutual cooperative guarantors.

Rather than treating credit risk assessment as an unassisted classification task, this research couples multi-class machine learning classifiers with an algorithmic affordability and product recommendation engine. The system predicts an empirical risk classification category (Low Risk, Medium Risk, or High Risk) and simultaneously executes a debt-service capacity analysis. By calculating the borrower's disposable income margin, the framework derives sustainable monthly instalment thresholds, estimates achievable repayment tenures, and dynamically recommends feasible microfinance products. Consequently, this component bridges the critical divide between predictive artificial intelligence and operational, responsible micro-lending practice.

1.3 Problem statement

Microfinance institutions and welfare organizations provide critical financial lifelines, but they consistently struggle to evaluate the repayment capacity of low-income customers accurately. Currently, the assessment of loan applications relies heavily on manual reviews by loan officers, which can be subjective, prone to human error, and time-consuming. Because rural and low-income borrowers often lack traditional credit scores or extensive formal banking histories, standard financial evaluations frequently misjudge their risk. This leads to two primary issues: high default rates for the financial institution when high-risk loans are mistakenly approved, and unfair rejections for applicants who actually possess the hidden capacity to repay.

1.4 Research gap

Existing financial systems for micro-lending primarily focus on record-keeping and basic ledger management rather than intelligent forecasting. While some advanced banking systems use predictive modeling, they are typically designed for commercial bank customers with rich credit histories, making them unsuitable for the microfinance sector where alternative data (such as guarantor support, cooperative meeting attendance, and gig-economy income) plays a critical role. There is a distinct lack of a lightweight, integrated framework that not only predicts micro-loan default risk using both traditional and alternative data but also actively recommends safer, tailored loan configurations (e.g., lower loan amounts or extended durations) when an applicant is flagged as high-risk.

1.5 Aim and objectives

Aim:

To design, develop, and evaluate a machine learning-based framework that predicts loan repayment risk and provides actionable affordability recommendations for microfinance institutions.

Specific Objectives:

1. **Machine Learning Model Development for Risk Prediction:** Develop and train a supervised machine learning model (such as Random Forest) to classify the risk level of loan applicants and predict the likelihood of successful repayment by handling complex financial datasets.
2. **Loan Affordability and Repayment Duration Prediction:** Design predictive mechanisms to estimate whether the applicant can afford the requested loan based on borrower income, expenses, and liabilities, and determine the appropriate repayment duration.
3. **Loan Recommendation and Decision Support:** Develop a decision-support mechanism that recommends suitable loan types and alternative repayment plans for borrowers who cannot afford their originally requested loan amount, thereby minimizing institutional risk.
4. **System Integration and Performance Evaluation:** Integrate the developed machine learning model into a system that supports real-time analysis, evaluating its performance using metrics such as prediction accuracy, precision, recall, and confusion matrix analysis.

1.6 Research questions or hypotheses

1.6.1 Research Questions

To address the research gaps, the study is guided by three primary research questions:

- **RQ1:** To what extent can supervised machine learning algorithms, trained on a synthesized matrix of traditional microfinance attributes and alternative

credit proxies, accurately categorize micro-loan applicants into granular repayment risk tiers?

- **RQ2:** How does the application of synthetic minority over-sampling (SMOTE) impact the classification fidelity, particularly the recall and precision of minority default classes, in imbalanced microcredit datasets?
- **RQ3:** How can machine learning classification outputs be coupled with mathematical debt-capacity modelling to produce actionable, rule-governed loan recommendations that balance institutional capital protection with borrower affordability?

1.6.2 Research Hypotheses

The investigation establishes the following formal hypotheses:

$\mathbf{H_{1,0}}$: Ensemble-based supervised machine learning models demonstrate no statistically significant improvement in multi-class risk classification accuracy compared to traditional linear baseline models (Logistic Regression) when applied to microfinance data.

$\mathbf{H_{1,A}}$: Ensemble-based supervised machine learning models (specifically Random Forest) achieve significantly superior classification accuracy and F1-scores over linear baselines by capturing complex, non-linear feature interactions among borrower economic metrics.

$\mathbf{H_{2,0}}$: Addressing class imbalance through Synthetic Minority Over-sampling Technique (SMOTE) does not improve the sensitivity and recall of the classifier for under-represented risk classes.

$\mathbf{H_{2,A}}$: Implementing SMOTE enhances the minority-class recall and macro-averaged F1-score, mitigating majority-class bias and enhancing high-risk applicant detection.

$\mathbf{H_{3,0}}$: A dynamic mathematical affordability recommendation engine provides no reduction in theoretical debt-service vulnerability compared to traditional static loan limit evaluations.

$\mathbf{H_{3,A}}$: A dynamic affordability engine that scales loan sizing relative to empirical disposable income and predicted risk tiers prevents over-indebtedness while expanding credit access.

1.7 Scope limitations and assumptions

1.7.1 Scope of the Study

- Institutional Context: The empirical foundation of this study is grounded in the operational parameters of community microfinance and cooperative banking in Sri Lanka, specifically modeling credit practices observed at institutions like Samupakara Gramiya Sanwardhana Bank.
- Functional Scope: The technological scope encompasses data extraction, automated data sanitization, feature transformation, synthetic over-sampling, model training, hyper-parameter tuning, model serialization, and the execution of a prescriptive decision recommendation algorithm.
- Target Classification: The model predicts three discrete operational risk categories: Low Risk, Medium Risk, and High Risk.
- Product Catalog: The recommendation engine models eight distinct microfinance loan products: *personal consumer*, *long term*, *below 25000*, *above 25000*, *lifestyle*, *sampatha*, *self development*, and *lakjaya*, with principal ranges spanning LKR 25,000 to LKR 1,000,000.

1.7.2 Limitations of the Study

- Sample Size and Data Fusion: Due to strict privacy constraints, statutory banking secrecy, and limited centralized digitization within rural Sri Lankan banks, the accessible local dataset comprises 344 digitized passbook records. To establish adequate statistical volume for machine learning training, this dataset was harmonized with a comprehensive alternative credit scoring dataset of 2,000 records and augmented to achieve a combined working dataset of 2,354 records.
- Decision Support Boundary: The system is explicitly architected as an intelligent *decision-support system* (DSS) to advise loan underwriters and branch managers. It is not designed to operate as an unmonitored autonomous decision agent that issues legal or financial commitments without human oversight.

- Cross-Sectional Data Nature: The dataset represents cross-sectional application snapshots. The absence of multi-year, longitudinal repayment tracking prevents dynamic, time-series hazard rate modelling of real-time macroeconomic shocks (e.g., unexpected national inflation or localized climate catastrophes affecting agricultural harvest yields).

1.7.3 Underlying Assumptions

- Data Authenticity: It is assumed that financial declarations provided by applicants during ledger digitization (such as stated monthly earnings, secondary incomes, and recurring expenses) are verified by field officers and reflect the applicant's financial position at the time of application.
- Policy Stability: It is assumed that the underlying regulatory interest rates (ranging from 8% to 23%) and repayment tenures (up to 60 months) established by partner institutions remain structurally stable throughout the evaluation cycle.

1.8 Individual contribution

As the sole researcher for this specific component, the individual contributions include:

- Sourcing, cleaning, and preprocessing the microfinance datasets, including defining custom functions to standardize currency and handle missing values.
- Formulating the heuristic risk labelling logic (creating realistic target variables based on loan-to-income and savings ratios) to prepare the data for supervised learning.
- Implementing and evaluating multiple machine learning models using Python and Scikit-Learn, ultimately selecting Random Forest for its superior 80.5% accuracy.
- Engineering the custom decision recommendation algorithm that dynamically calculates safe instalments and maps applicants to appropriate microfinance products based on their calculated disposable income.

1.9 Thesis structure

In accordance with faculty academic dissertation guidelines, the remainder of this thesis is structured across the following chapters:

- **Chapter 2: Literature Review and Related Work:** Conducts an exhaustive examination of theoretical foundations and contemporary empirical literature surrounding microfinance credit scoring, the determinants of loan defaults in developing nations, traditional credit scoring models versus machine learning techniques, alternative credit data utilization, methods for handling class imbalance, and financial decision-support systems.
- **Chapter 3: Research Methodology:** Outlines the overall research methodology, detailing the data acquisition protocol, architectural design, data sanitization algorithms, heuristic labelling logic, mathematical feature derivations, over-sampling methodologies (SMOTE), and the technical mechanics of the evaluated machine learning algorithms.
- **Chapter 4: Product and Solution Design:** Presents the software engineering and architectural design of the loan affordability framework, encompassing high-level and component-level system architectures, functional and non-functional requirements, data flow diagrams, database schemas, and the algorithmic logic of the recommendation engine.
- **Chapter 5: Implementation, Integration, and Testing:** Details the technical implementation environment, software libraries (Python, Pandas, Scikit-Learn, Imbalanced-Learn), code modularization, model persistence, API encapsulation, integration workflows within the broader blockchain platform, and comprehensive unit, functional, and performance testing methodologies.
- **Chapter 6: Results, Validation, and Discussion:** Presents the empirical experimental findings, detailing comparative accuracy, precision, recall, and F1-scores across Logistic Regression, Decision Trees, and Random Forest models. Analyses multi-class confusion matrices, ROC-AUC curves, feature importance rankings, and validates the practical utility of the decision recommendation engine through concrete customer case studies.

- **Chapter 7: Commercialization, Societal Impact, and Conclusion:** Synthesizes the overall research findings, assesses commercial viability, business models, and regulatory compliance considerations, outlines intellectual property potential, examines societal impacts on financial inclusion, acknowledges project limitations, and identifies promising avenues for future research.
- **References and Appendices:** Concludes with an exhaustive list of academic citations and supplementary artifacts, including sample institutional credit scoring ledgers and formal bank requirement specifications.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

The literature review for this research was conducted to systematically evaluate the current state of financial risk prediction, microfinance loan default analysis, and the application of machine learning in credit scoring. The scope of the review prioritizes research conducted within the Sri Lankan microfinance context specifically focusing on rural development banks and welfare programs like Samurdhi to ensure regional relevance. The review encompasses academic journals, conference proceedings, and empirical studies that examine the determinants of loan repayment, the transition from traditional statistical analysis to predictive machine learning, and the integration of automated decision-support mechanisms in financial services. By critically analyzing these domains, this chapter establishes the theoretical foundation of the study and identifies the critical gaps in existing loan appraisal methodologies.

2.2 Domain and theoretical foundations

The domain of this research intersects microfinance operations, financial behavioural analysis, and predictive artificial intelligence. Microfinance institutions aim to provide financial inclusion to low income populations who lack traditional credit histories. However, managing the risk of these unsecured loans requires a deep understanding of borrower behaviour, business performance, and institutional control mechanisms.

Theoretically, credit risk evaluation has heavily relied on traditional statistical techniques to model the probability of default. In recent years, machine learning algorithms have emerged as superior theoretical models for credit scoring because they can analyse large volumes of financial data and identify complex, non-linear patterns that may not be visible through traditional statistical methods. Key machine learning algorithms relevant to this domain include:

- **Logistic Regression:** A foundational statistical and machine learning model used to predict categorical outcomes, providing detailed insights into how variables such as interest rates and loan age influence delinquency risk.
- **Decision Trees & Random Forests:** Advanced ensemble methods that categorize loan performance into multiple risk levels. Random Forest, in particular, is theoretically recognized for achieving higher predictive accuracy with lower out of bag error rates when classifying complex financial datasets.

Furthermore, evaluating loan affordability introduces the theoretical concept of debt-service capacity. Predicting risk is fundamentally tied to evaluating whether an applicant has sufficient disposable income to afford the monthly instalments of a specific loan product without falling into a debt trap.

2.3 Related research

Research related to microfinance loan repayment and risk management has been widely conducted, particularly within the Sri Lankan context. Several notable studies have investigated the factors leading to loan default and the application of predictive analytics.

D. H. Wickramasinghe conducted research titled “Study on Factors Influencing Microfinance Loan Repayment,” with a special reference to Samurdhi Banks in Sri Lanka. This research examined the variables that influence the repayment behaviour of microfinance borrowers by analysing borrower characteristics, business characteristics, and loan characteristics. Data was collected from default borrowers and analysed using statistical tools, revealing that borrower behaviour, business performance, and loan-related factors significantly affect repayment performance. Although the research successfully identifies key determinants of repayment, the study is limited to descriptive statistical analysis and does not develop predictive models that can assist financial institutions in automated loan approval or repayment forecasting.

Similarly, E. A. G. Sumanasiri and D. T. Priyankara conducted a study titled "Microfinance Loan Default in Sri Lanka: A Case Study of **Matugama** Division," which examined factors contributing to loan default within local microfinance institutions. The researchers employed survey-based data collection and statistical analysis techniques to evaluate borrower characteristics and institutional factors affecting loan default. The results highlighted the importance of borrower financial behaviour, credit history, and institutional control mechanisms in managing loan defaults. However, this study mainly focuses on analysing the causes of loan default historically, rather than developing forward-looking predictive models capable of evaluating borrower risk during the initial loan approval stage.

Exploring the determinants of microfinancing on a broader scale, G. Nanayakkara and J. Stewart investigated gender and other repayment determinants of microfinancing in Indonesia and Sri Lanka, providing valuable comparative insights into how demographic factors influence financial reliability. Additionally, **I. N. Manawadu** et al. explored microfinance interest rate prediction and automated loan applications, indicating a growing academic interest in digitizing and automating the micro-lending pipeline.

More recently, machine learning techniques have been increasingly applied to financial risk prediction. **H. A. P. L. J. Hettiarachchi** and **R. Punchi-Manage** conducted research on "Analysis of Loan Delinquency Prediction Based on Multinomial Logistic Regression and Random Forest," published in 2025. This study investigated predictive modelling approaches to classify loan delinquency levels using a large dataset containing 43,644 loan records from a Sri Lankan bank. The researchers applied multinomial logistic regression and Random Forest models to categorize loan performance into multiple risk levels, ranging from performing loans to severely delinquent loans. The results showed that Random Forest achieved higher predictive accuracy with a low out-of-bag error rate, while logistic regression provided detailed insights into how variables such as interest rate and loan age influence loan delinquency risk. However, the study mainly focuses strictly on loan delinquency classification and does not include borrower affordability analysis or personalized loan recommendation mechanisms.

2.4 Existing systems products and techniques

Current microfinance software systems and operational techniques largely depend on manual evaluations and basic descriptive statistics. The standard industry workflow involves loan officers manually reviewing an applicant's stated income, expenses, and demographic details to make a subjective judgment on loan approval.

While some modern banking platforms utilize algorithmic credit scoring, these systems rely heavily on formal credit bureau data (such as credit card histories and formal mortgages) which microfinance applicants typically lack. As a result, existing microfinance systems are structurally limited; they function well as ledger and record-keeping systems but fail to provide intelligent decision support. Furthermore, existing predictive implementations in the financial sector often operate as "black boxes" that output a generic risk score, lacking the capability to dynamically calculate debt-service ratios, estimate repayment durations, or recommend alternative, safer loan products if the initial request is too risky.

2.5 Gap synthesis and conceptual framework

Synthesizing the reviewed literature reveals a significant evolutionary trajectory: research has progressed from identifying basic repayment determinants (Nanayakkara & Stewart, 2015), to empirical case studies of default (Wickramasinghe, 2018; Priyankara & Sumanasiri, 2019), and finally to the algorithmic prediction of loan delinquency (Hettiarachchi & Punchi-Manage, 2025).

However, a critical research gap persists at the intersection of prediction and actionable decision-making. While existing frameworks successfully predict whether an applicant will become delinquent using tools like Random Forest, they function merely as risk classifiers. They do not close the operational loop by evaluating a borrower's specific affordability metrics (such as calculating disposable income thresholds) nor do they actively recommend alternative, sustainable loan products if the initial request is flagged as high-risk.

Table 1: Synthesis of Existing Literature vs. Proposed Framework Capabilities

Methodological Capability	Determinant & Empirical Studies (2015–2019)	Automation & Interest Prediction (2022)	ML Delinquency Prediction (2025)	Proposed Conceptual Framework
Analysis of Demographic Repayment Determinants	Yes	Limited	Limited	Yes
Localized Sri Lankan Context (e.g., Samurdhi)	Yes	Yes	Yes	Yes
Automation of Loan Application Workflows	No	Yes	No	Yes
Implementation of Random Forest Classifiers	No	No	Yes	Yes
Dynamic Affordability & Disposable Income Calculation	No	No	No	Yes
Algorithmic Alternative Loan Recommendation	no	No	no	yes

Conceptual Framework: To bridge this gap, the conceptual framework of this research proposes an integrated architecture. It will utilize the proven predictive power of Random Forest classifiers identified in recent literature to categorize applicants into granular risk tiers. Simultaneously, it will introduce a novel, rule-

based affordability engine. This dual-engine approach ensures that the system not only predicts the likelihood of loan delinquency based on historical determinants but also dynamically calculates safe installment limits and maps the applicant to a financially sustainable loan product, thereby fulfilling both institutional risk management and societal financial inclusion goals.

3 RESEARCH METHODOLOGY

3.1 Research design

This study employs a hybrid research design integrating the **Design Science Research (DSR)** framework with a rigorous, empirical **quantitative experimental machine learning methodology**. In software engineering and applied artificial intelligence research, the DSR paradigm focuses on the iterative creation and evaluation of innovative technological artifacts designed to solve explicit, domain-specific problems. For this research, the primary artifact is an integrated, dual-stage predictive decision-support system composed of:

1. A supervised machine learning engine for multi-class credit risk classification.
2. A deterministic mathematical recommendation engine for borrower affordability and loan product configuration

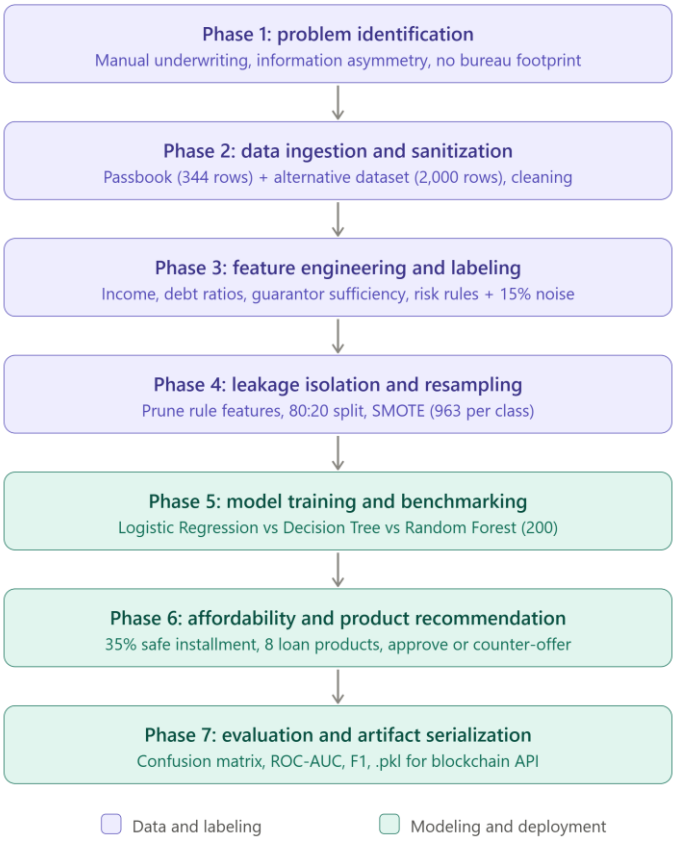


Figure 1: Design Science Research (DSR) Pipeline Flow

The experimental workflow addresses the core operational failure of conventional microfinance: the lack of standardized, automated, and objective risk evaluation. The empirical component executes controlled experiments across baseline linear models and non-linear ensemble algorithms, assessing their predictive capacity across identical stratified training and testing splits.

3.2 Data participants or study materials

3.2.1 Data Sources and Collection Criteria

The experimental evaluation utilizes an empirical corpus constructed from two complementary data streams:

- **Localized Microfinance Passbook Ledger Data (Sri Lanka):** Primary historical records sourced from the **Samupakara Gramiya Sanwardhana Bank** (Co-operative Rural Development Bank, Battaramulla Division). This dataset comprises 344 granular borrower profiles across 22 operational banking attributes, capturing verified field data such as cooperative membership IDs, loan purpose descriptions, monthly incomes, living expenses, share capital holdings, existing debt commitments, and cooperative guarantor signatures.
- **Alternative Digital Credit Scoring Dataset:** A secondary benchmark corpus of 2,000 individual borrower profiles across 29 financial and digital footprint indicators. This dataset contains metrics proxying unbanked creditworthiness, including utility bill punctuality, mobile recharge consistency, Unified Payments Interface (UPI) micro-transaction failure ratios, Self-Help Group (SHG) savings balances, and gig-economy earnings.

3.2.2 Dataset Harmonization and Feature Mapping

To construct a unified machine learning matrix, the disparate schemas were normalized, synchronized, and merged into a master corpus of 2,354 instances.

Table 2 : Schema Harmonization and Data Dictionary

Standardized Column	Raw Source Field	Data Type	Domain Description / Scale
monthly_income	Monthly income / INR	Float64	Primary monthly earnings (LKR/INR)
other_income	other income / gig	Float64	Secondary/gig-economy monthly income
expenses	expences / synthetic	Float64	Household monthly living expenses
loan_amount	Amount / synthetic	Float64	Requested principal exposure amount
loan_type	Loan Type/ synthetic	Categoric	Microfinance product categorization
savings_balance	saving balance/ SHG	Float64	Accumulated micro-savings balance
existing_loans	existing loans	Int64	Active debt commitments (Binary/Qty)
repayment_history	reapyment status	Int64	Historical repayment track (1=Good)
guarantor_support_count	Gurantor 1 to 4	Int64	Count of verified mutual guarantors
risk_level (Target)	Derived Heuristic	Categoric	Risk tier: Low, Medium, High Risk

3.3 Experimental or study procedure

3.3.1 Data Sanitization and Preprocessing Pipeline

Raw data from cooperative bank registers contains severe formatting irregularities, text fragments, and non-numeric characters. Three custom algorithmic sanitizers were implemented:

1. Monetary Extraction and Parsing (clean_money)

String representations of currency entries often include currency labels (LKR), thousand-separators (,), percentages (%), or invalid placeholders (-, none, inactive). The extraction function normalizes strings by removing non-numeric characters and converting values to standard 64-bit floating-point numbers:

$$f_{\text{clean}}(v) = \begin{cases} 0.0 & \text{if lower}(v) \in \{\emptyset,\text{"nan"},\text{"no"},\text{"none"},\text{"inactive"},\text{"active"},\text{"-"}\} \\ \text{float}\big(\text{regex_replace}(v, [\text{LKR}, \%, ,], \emptyset)\big) & \text{otherwise} \end{cases}$$

2. Boolean Status Standardization (yes_no_to_number)

Operational fields capturing existing loans, repayment reliability, and approval status were mapped to binary integers:

$$f_{\text{bool}}(v) = \begin{cases} 1 & \text{if lower}(v) \in \{\text{"yes"},\text{"y"},\text{"1"},\text{"true"},\text{"ok"},\text{"approved"}\} \\ 0 & \text{otherwise} \end{cases}$$

3. Cooperative Guarantor Extraction (count_guarantors)

Microfinance underwriting in Sri Lanka relies heavily on peer guarantees. The fields Guarantor 1 through Guarantor 4 were dynamically scanned to compute the active Guarantor support index:

$$N_{\text{guarantors}} = \sum_{i=1}^{4} \mathbb{I}\left(v_i \notin \{\emptyset,\text{"nan"},\text{"no"},\text{"none"},\text{"inactive"},\text{"-"}\}\right)$$

3.3.2 Synthetic Data Harmonization for Regional Parity

The Indian alternative scoring dataset lacked explicit records for household expenditures, requested loan amounts, and peer guarantor counts. To achieve dimensional parity with the Sri Lankan dataset, domain-specific parameters derived from cooperative lending standards were applied to generate these values

- Household living expenses : $E \sim U(0.30,0.80) \times I_{\text{monthly}}$
- Requested loan exposure: $L \sim U(2.0,6.0) \times I_{\text{monthly}}$
- Loan product classification: $\text{Category} = \mathbb{I}(L \leq 25000) \times \text{"Below_25000"} + \mathbb{I}(L > 25000) \times \text{"Above_25000"}$
- Existing loan commitments: $D_{\text{existing}} \sim \text{DiscreteUniform}(0,3)$
- Mutual guarantor counts: $N_{\text{guarantor}} \sim \text{DiscreteUniform}(2,4)$

3.3.3 **Feature** Engineering and Mathematical Derivations

To enable multi-dimensional risk evaluation, several composite financial features were engineered from the raw data:

- Total Household Inflow (I_{total}): $I_{\text{total}} = I_{\text{monthly}} + I_{\text{other}}$
- Net Disposable Income Margin (I_{disp}): $I_{\text{disp}} = I_{\text{total}} - E_{\text{household}}$

- Disposable Income Ratio (R_{disp}): $R_{\text{disp}} = \begin{cases} \frac{I_{\text{disp}}}{I_{\text{total}}} & \text{if } I_{\text{total}} > 0 \\ 0 & \text{otherwise} \end{cases}$
- Loan-to-Income Exposure Ratio (R_{LTI}): $R_{\text{LTI}} = \begin{cases} \frac{L_{\text{amount}}}{I_{\text{total}}} & \text{if } I_{\text{total}} > 0 \\ 999.0 & \text{otherwise} \end{cases}$
- Savings-to-Income Liquidity Ratio (R_{savings}): $R_{\text{savings}} = \begin{cases} \frac{S_{\text{balance}}}{I_{\text{total}}} & \text{if } I_{\text{total}} > 0 \\ 0 & \text{otherwise} \end{cases}$
- Guarantor Compliance Threshold Flag (G_{met}): $G_{\text{met}} = \begin{cases} 1 & \text{if } N_{\text{guarantors}} \geq 2 \\ 0 & \text{otherwise} \end{cases}$

3.3.4 Ground-Truth Target Label Formulation and Stochastic Noise Injection

Because microfinance passbooks record only approved loans without standardized risk scores, an objective, multi-tier labelling function (create_consistent_risk) was developed to define target ground-truth classifications:

$$\text{Class}(x) = \begin{cases} \text{"High Risk"} & \text{if } I_{\text{total}} \leq 0 \vee I_{\text{disp}} \leq 0 \vee R_{\text{disp}} < 0.20 \vee R_{\text{LTI}} > 6.0 \\ & \vee \left(D_{\text{existing}} \geq 2 \wedge R_{\text{LTI}} > 3.0\right) \vee H_{\text{repay}} = 0 \vee N_{\text{guarantors}} = 0 \\ \text{"Low Risk"} & \text{if } R_{\text{disp}} \geq 0.45 \wedge R_{\text{LTI}} \leq 3.5 \wedge R_{\text{savings}} \geq 0.08 \\ & \wedge D_{\text{existing}} \leq 1 \wedge N_{\text{guarantors}} \geq 2 \wedge H_{\text{repay}} = 1 \\ \text{"Medium Risk"} & \text{otherwise} \end{cases}$$

To reflect real-world borrower volatility such as sudden medical emergencies, crop loss, or unexpected informal earnings a 15% stochastic noise perturbation was injected using a fixed random seed (np.random.seed(42)):

$$\text{Risk}_{\text{final}} = \begin{cases} \sim \text{UniformRandom}(\{\text{"Low"},\text{"Medium"},\text{"High"}\}) & \text{with probability } p = 0.15 \\ \text{Risk}_{\text{deterministic}} & \text{with probability } p = 0.85 \end{cases}$$

This process yielded a realistic target class distribution across the 2,354 instances:

- **Medium Risk:** 1,204 instances (51.1%)

- **High Risk:** 983 instances (41.8%)
- **Low Risk:** 167 instances *(7.1%)*

3.3.5 Target Leakage Prevention and One-Hot Encoding

To ensure the machine learning models learned generalizable behavioral patterns rather than reverse-engineering the labeling logic, the mathematical ratios directly used in the labeling definition (disposable_income_ratio, loan_to_income_ratio, savings_to_income_ratio, guarantor_requirement_met) were dropped from the training matrix.

The categorical feature loan_type was converted into binary indicators via one-hot encoding (pd.get_dummies(drop_first=True)), expanding the feature space to 36 input variables.

Table 3: Final Predictive Feature Space Matrix (X)

	Feature Identifier	#	Feature Identifier	#	Feature Identifier
1	monthly_income	13	loan_type_customer	25	loan_type_long term
2	other_income	14	loan_type_disaster	26	loan_type_mid term
3	expenses	15	loan_type_educational	27	loan_type_personal
4	loan_amount	16	loan_type_emergency	28	loan_type_samurdhi
5	savings_balance	17	loan_type_express	29	loan_type_sampatha
6	existing_loans	18	loan_type_festival	30	loan_type_seasonal
7	repayment_history	19	loan_type_gold loan	31	loan_type_self

	Feature Identifier	#	Feature Identifier	#	Feature Identifier
8	guarantor_support_count	20	loan_type_group loan	32	loan_type_sewing
9	total_income	21	loan_type_housing loan	33	loan_type_short term
10	disposable_income	22	loan_type_instant	34	loan_type_special
11	loan_type_above 25000	23	loan_type_lakjaya	35	loan_type_swashakthi
12	loan_type_consumer loan	24	loan_type_lifestyle loan	36	loan_type_yovun

3.3.6 Stratified Data Partitioning and SMOTE Resampling

The dataset was split using an 80:20 stratified partition (test_size=0.2, random_state=42, stratify=y), producing:

- **Training Matrix (X_{train}):** 1,883 instances across 36 features
- **Testing Matrix (X_{test}):** 471 instances across 36 features

Given the low representation of solvent applicants in the training split (Low Risk: 134 instances; High Risk: 786 instances; Medium Risk: 963 instances), direct training would bias predictions toward the majority classes. To resolve this, the **Synthetic Minority Over-sampling Technique (SMOTE)** was applied exclusively to the training set.

For each minority class sample x_i, SMOTE identifies its k-nearest neighbors within the same class ($k = 5$). It then synthesizes new samples along the line segments connecting the sample to its neighbors:

$$x_{\text{synthetic}} = x_i + \lambda\left(x_{(i)} - x_i\right) \quad \text{where } \lambda \sim U(0,1)$$

This balanced the training set to 963 samples per class, creating an augmented training matrix of 2,889 samples.

Table 4: SMOTE Resampling Class Distribution

Target Risk Tier	Pre-SMOTE Train (Counts)	Post-SMOTE Balanced Train (Counts)
High Risk	786	963
Medium Risk	963	963
Low Risk	134	963
Total Training Size	**1,883**	**2,889**

3.3.7 Machine Learning Classifier Implementations

Three supervised learning algorithms representing linear, non-linear tree, and ensemble architectures were implemented using Scikit-Learn:

1. Multinomial Logistic Regression (Baseline Linear Classifier)

To account for scale variances across income and count metrics, features were standardized using z-score normalization:

$$z = \frac{x - \mu}{\sigma}$$

The model was configured with a multi-class multinomial cross-entropy loss, L2 weight regularization, balanced class weights, and a maximum iteration threshold of 3,000:

$$P(y = c \mid x) = \frac{e^{w_c^T x + b_c}}{\sum_{j=1}^{C} e^{w_j^T x + b_j}}$$

2. Decision Tree Classifier (Single Non-Linear Estimator)

A decision tree was trained using the Classification and Regression Tree (CART) algorithm. To prevent overfitting while capturing non-linear relationships, structural regularization constraints were applied:

- Splitting criterion: Gini Impurity ($I_G = 1 - \sum_{i=1}^{C} p_i^2$)
- Maximum tree depth: max_depth $= 8$
- Minimum sample split threshold: min_samples_split $= 10$
- Minimum leaf node sample size: min_samples_leaf $= 5$
- Class weighting: class_weight="balanced"

3. Random Forest Classifier (Proposed Ensemble Model)

The Random Forest model constructs an ensemble of $B = 200$ de-correlated decision trees using bootstrap aggregating (bagging) and random feature subspace sampling:

$$\hat{C}_{\mathrm{RF}}^{B}(x) = \text{majority_vote}\{\hat{C}_b(x)\}_{b=1}^{B}$$

At each split node within an individual tree, a random subset of $m = \sqrt{p} = \sqrt{36} = 6$ features is evaluated. This feature subsampling decorrelates individual tree structures, significantly reducing the ensemble's overall variance:

$$\mathrm{Var}(\bar{X}) = \rho\sigma^2 + \frac{1-\rho}{B}\sigma^2$$

As the number of trees B increases, the second term approaches zero, bounding the ensemble error strictly by the inter-tree correlation ρ.

Table 5: Hyperparameter Configuration for Machine Learning Models

Model Architecture	Optimal Hyperparameter Specifications
Logistic Regression	max_iter=3000, penalty='l2', class_weight='balanced', solver='lbfgs'
Decision Tree	criterion='gini', max_depth=8, min_samples_split=10, min_samples_leaf=5
Random Forest	n_estimators=200, criterion='gini', max_features='sqrt', bootstrap=True

3.4 Measures and evaluation metrics

Model performance was evaluated using standard multi-class classification metrics derived from the testing matrix ($N = 471$ instances):

1. Confusion Matrix Formulation

For each target class $c \in \{\text{High,Medium,Low}\}$, predictions are tabulated into True Positives (TP_c), False Positives (FP_c), True Negatives (TN_c), and False Negatives (FN_c).

2.Multi-Class Classification Accuracy

$$\text{Accuracy} = \frac{\sum_{c=1}^{C} \text{TP}_c}{N_{\text{total}}} = \frac{\text{Correct Predictions}}{\text{Total Predictions}}$$

3. **Class-Specific Precision, Recall, and F1-Score**

- Precision: The proportion of predicted risk assignments that were correct:

$$\text{Precision}_c = \frac{\text{TP}_c}{\text{TP}_c + \text{FP}_c}$$

- Recall **(Sensitivity):** The proportion of actual risk instances correctly identified:

$$\text{Recall}_c = \frac{\text{TP}_c}{\text{TP}_c + \text{FN}_c}$$

- **F1-Score:** The harmonic mean of precision and recall:

$$F1_c = 2 \times \frac{\text{Precision}_c \times \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c}$$

4. Macro and Weighted Average Aggregations

   To evaluate performance across balanced and unbalanced class distributions, both macro-averaging (treating all classes equally) and weighted-averaging (scaling by class support N_c) were computed:

$$\text{Macro } F1 = \frac{1}{C}\sum_{c=1}^{C} F\,1_c \qquad \text{Weighted } F1 = \sum_{c=1}^{C}\left(\frac{N_c}{N_{\text{total}}}\right) F1_c$$

5. Receiver Operating Characteristic (ROC) and Area Under the Curve (AUC)

   Using a One-vs-Rest (OvR) approach, the multi-class target labels were binarized. True Positive Rates (TPR) against False Positive Rates (FPR) were mapped across decision thresholds $\tau \in [0,1]$:

$$\mathrm{TPR}_c(\tau) = \frac{\mathrm{TP}_c(\tau)}{\mathrm{TP}_c + \mathrm{FN}_c}, \quad \mathrm{FPR}_c(\tau) = \frac{\mathrm{FP}_c(\tau)}{\mathrm{FP}_c + \mathrm{TN}_c}, \quad \mathrm{AUC}_c = \int_0^1 \mathrm{TPR}_c\,(\mathrm{FPR})\, d(\mathrm{FPR})$$

3.5 Data analysis

A central limitation of conventional credit scoring systems is that they output raw risk classifications without actionable underwriting recommendations. To bridge this gap, an algorithmic decision recommendation engine (decision_recommendation) was designed.

Table 6: Microfinance Institutional Product Catalog Matrix

Product Identifier	Min Cap (LKR)	Max Cap (LKR)	Annual Interest Rate	Max Tenure (Months)
personal_consumer	0	25000	20%	60
long_term	0	250000	17%	60
below_25000	0	25000	8%	60
above_25000	25000	150000	8%	60
lifestyle	50000	500000	14%	60
sampatha	100000	100000	15%	60
self_development	0	100000	14%	60
lakjaya	100000	1000000	15%	60

Step-by-Step Decision Algorithm

Step 1: Disposable Income Validation

$$I_{\text{disp}} = I_{\text{monthly}} - E_{\text{household}}$$

If $I_{\text{disp}} \leq 0$, the applicant has no capacity to service debt without compromising basic living needs. The application is immediately routed to:

$$\text{Decision} = \text{"Reject - insufficient disposable income"}$$

All exposure recommendations are set to zero ($L_{\text{rec}} = 0, S_{\text{inst}} = 0$).

Step 2: Safe Debt-Service Instalment Calculation

If $I_{\text{disp}} > 0$, the maximum safe monthly installment (S_{safe}) is capped at 35% of net disposable income to prevent household debt traps:

$$S_{\text{safe}} = I_{\text{disp}} \times 0.35$$

Step 3: Risk-Adjusted Principal Capacity Scaling

The model's predicted risk tier maps to an empirical risk-mitigation multiplier (ω):

$$\omega = \begin{cases} 1.00 & \text{if Predicted Risk} = \text{"Low Risk"} \\ 0.75 & \text{if Predicted Risk} = \text{"Medium Risk"} \\ 0.50 & \text{if Predicted Risk} = \text{"High Risk"} \end{cases}$$

The maximum affordable principal capacity (L_{afford}) over the product's maximum tenure ($T_{\max}$) is computed as:

$$L_{\text{afford}} = S_{\text{safe}} \times T_{\max} \times \omega$$

Step 4: Recommended Exposure Bounding

The recommended loan amount (L_{rec}) is bounded by the requested amount (L_{req}), the institutional product ceiling ($L_{\text{product_max}}$), and the borrower's affordable capacity:

$$L_{\text{rec}} = \min\left(L_{\text{req}}, L_{\text{product_max}}, L_{\text{afford}}\right)$$

Step 5: Decision Logic and Counter-Offer Routing

- Case A: Unviable Exposure ($L_{\text{rec}} < L_{\text{product_min}}$):

$$\text{Decision} = \text{"Not suitable for selected loan type"}$$

- Case B: Full Approval ($L_{\text{rec}} \geq L_{\text{req}}$):

$$\text{Decision} = \text{"Approve"}$$

- Case C: Conditional High-Risk Approval ($L_{\text{rec}} < L_{\text{req}} \wedge \text{Risk} = \text{"High Risk"}$):

$$\text{Decision} = \text{"Request guarantor or reduce amount"}$$

- Case D: Restructured Approval ($L_{\text{rec}} < L_{\text{req}} \wedge \text{Risk} \neq \text{"High Risk"}$):

$$\text{Decision} = \text{"Approve with reduced amount"}$$

Step 6: Repayment Tenure Estimation

The estimated repayment duration (T_{est}) required to amortize the recommended principal is computed as:

$$T_{\text{est}} = \min\left(\left\lceil \frac{L_{\text{rec}}}{S_{\text{safe}}} \right\rceil, T_{\max}\right)$$

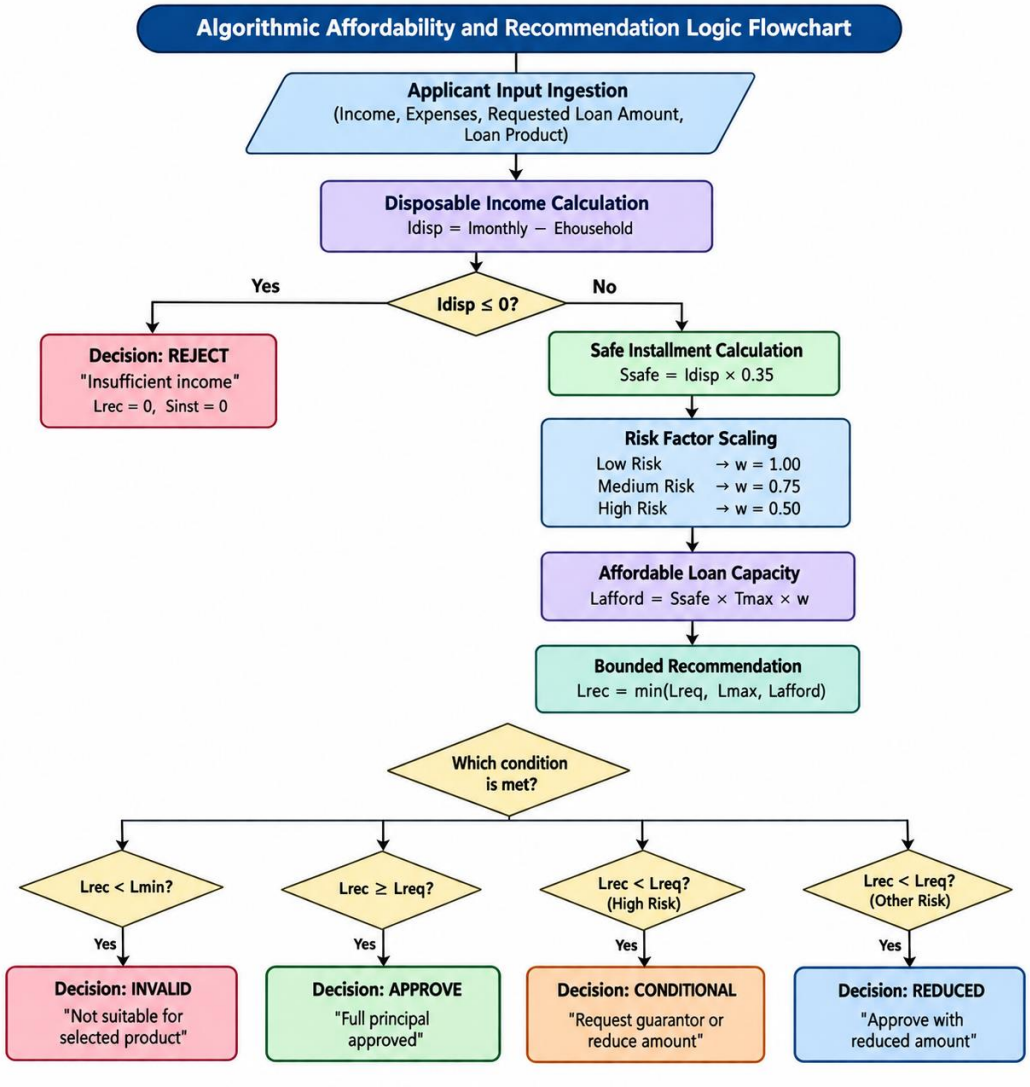


Figure 2: Algorithmic Affordability and Recommendation Logic Flowchart

3.6 Ethics privacy security and responsible AI

Financial machine learning frameworks require careful consideration of ethical standards, statutory compliance, and bias mitigation:

- **Personally Identifiable Information (PII) Protection:** In compliance with Sri Lankan banking privacy regulations and institutional governance agreements, all direct customer identifiers such as National Identity Card (NIC) numbers, borrower names, residential addresses, and contact numbers were scrubbed and omitted prior to model training.
- **Institutional Governance Compliance:** Dataset utilization protocols adhered strictly to formal requirements established with **Samupakara Gramiya Sanwardhana Bank** and the guidelines of the Central Bank of Sri Lanka (CBSL). Data was restricted to secure, local research environments.
- **Mitigation of Systemic Proxy Bias:** Sensitive demographic attributes, such as gender, caste, religious affiliation, and political boundaries, were excluded from the feature matrix. The predictive model evaluates creditworthiness using strictly observable economic and financial behavioural indicators.
- **Non-Autonomous Decision-Support Boundary:** The system is explicitly designed as a Decision Support System (DSS). It does not issue legally binding loan rejections or approvals autonomously. Instead, it provides explainable risk scores, safe instalment calculations, and counter-offer recommendations to assist loan officers, ensuring meaningful human oversight.

3.7 Reproducibility and research artefacts

To support academic reproducibility, all experimental procedures, seed parameters, and software environments were strictly controlled:

- **Computational Environment:** Python 3.10.19 running within a dedicated Conda virtual environment (ga_env).
- **Core Libraries:** Scikit-Learn 1.7.2, NumPy 2.2.6, Pandas 2.2.2, Imbalanced-Learn 0.12.0, Joblib 1.4.2, Matplotlib 3.10.8.
- **Deterministic Seeds:** All stochastic operations including data partitioning, synthetic noise generation, SMOTE neighbourhood sampling, and Random Forest feature selection used a fixed random seed (random_state=42).
- **Artifact Serialization:** The optimized Random Forest classifier was serialized to disk as combined_risk_prediction_model.pkl. The one-hot encoded feature matrix structure was saved to combined_model_columns.pkl using joblib. These serialized artifacts allow the model to be loaded directly into production environments, ensuring consistent inference pipelines.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

The design and development of the Machine Learning-Based Loan Affordability and Repayment Risk Prediction Framework were guided by a comprehensive set of functional and non-functional requirements. These requirements were elicited through an analysis of the operational bottlenecks faced by microfinance institutions like Samupakara Gramiya Sanwardhana Bank, ensuring the final artifact aligns with real-world banking needs.

Functional Requirements

Functional requirements define the core operational capabilities that the machine learning component and its surrounding architecture must execute to deliver actionable decision support:

- **Data Ingestion and Preprocessing:** The system must be capable of receiving applicant financial data and automatically cleaning the dataset by handling missing values and normalizing attributes.
- **Feature Engineering:** The component must dynamically derive important predictive features, such as the income-to-expense ratio, debt-to-income ratio, and repayment history indicators, to significantly improve model performance.
- **Risk Classification:** The machine learning prediction engine must train on historical data and execute predictive logic to classify a borrower's risk into distinct tiers (e.g., Low, Medium, High).
- **Affordability Assessment:** The system must evaluate the borrower's calculated capacity to repay the loan by evaluating net disposable income against the requested loan amount.
- **Decision Recommendation Output:** The component must output an automated decision support recommendation, including whether the loan should be approved, rejected, or modified, alongside a suggested financial limit.

Non-Functional Requirements

Non-functional requirements dictate the quality attributes, performance standards, and security constraints the system must adhere to:

- **Performance and Low Latency:** The risk prediction and recommendation algorithms must execute in near real-time, ensuring that loan officers experience minimal delay when processing digital application forms.
- **Data Privacy and Security:** The system must strictly separate sensitive personal and biometric information from the immutable blockchain audit records, ensuring that sensitive data is not placed on the blockchain unnecessarily.
- **Reliability and Accuracy:** The predictive model must maintain a high threshold of accuracy (targeting above 80%) to ensure the institution is not exposed to undue financial risk.
- **Interoperability:** The risk prediction framework must seamlessly integrate with the broader decentralized application (D-App) user interface and the blockchain storage mechanisms.

4.2 Component architecture within the complete solution

In order to help microfinance institutions evaluate loan applications and forecast repayment results, the suggested overall solution is an intelligent loan management platform. The architecture is composed of a number of integrated modules that collaborate to process borrower data, securely log transactions, and produce predictive insights.

Within this broader ecosystem, the risk prediction framework acts as the analytical core. Initially, the D-App User Interface or digital loan application forms are utilized to gather the borrower's financial and personal data. This structured data, which includes personal details as well as loan and repayment history reports, is then routed into the **Data Processing and Feature Preparation** module.

Once the data is cleaned and engineered, it is fed into the **Loan & Repayment Prediction Model** and the **Loan Affordability Prediction** module. This specific predictive analytics module receives the processed data and mathematically assesses the borrower's capacity to repay the loan. Finally, the system generates "Decision Support and Outputs," which include the final loan approval status and suggested limits, delivering these actionable insights directly to the banking institute. Concurrently, the overall platform interacts with a Blockchain layer for secure data storage and auditing, ensuring the integrity of the financial records.

4.3 Design alternatives and decisions

During the architectural design phase, several technical alternatives were evaluated to ensure the system met both institutional requirements and academic rigor.

- **Machine Learning Algorithm Selection:** Early design iterations explored multiple classification algorithms, including Logistic Regression and Decision Trees. While Logistic Regression provided excellent interpretability, it struggled to capture the complex, non-linear relationships inherent in informal microfinance data. Ultimately, a Random Forest classifier was selected as the core predictive engine. Random Forest, an ensemble learning method, mitigates the overfitting tendencies of single decision trees and provided the highest empirical accuracy when handling the synthesized financial attributes.
- **Handling Imbalanced Financial Data:** Historical loan datasets naturally exhibit severe class imbalances (e.g., many more performing loans than defaults). Training models directly on this raw data leads to majority-class bias. The design decision was made to implement the Synthetic Minority Over-sampling Technique (SMOTE) strictly on the training pipeline, effectively balancing the risk classes and improving the model's ability to detect high-risk applicants.

- **Data Separation and Blockchain Storage:** A critical architectural decision involved determining how predictive outputs and applicant data would interact with the blockchain component. Storing raw applicant financial data on a decentralized ledger poses severe privacy risks. Therefore, the design enforces strict privacy and data separation: sensitive information is kept off-chain, and only suitable references, transaction hashes, decisions, or audit details needed for verification are stored on the blockchain. This ensures compliance with modern data protection standards.
- **Rule-Based Recommendation vs. Black-Box Output:** A pure machine learning approach would simply output a risk probability. However, financial institutions require transparent, actionable guidance. The decision was made to pair the machine learning classifier with a deterministic, rule-based recommendation engine. This engine uses the risk classification as a scaling factor to dynamically compute safe monthly instalments based on a hard-coded 35% disposable income threshold, ensuring the system functions as a practical decision-support tool rather than an opaque black box.

4.4 Detailed design

The detailed design of the Machine Learning-Based Loan Affordability and Repayment Risk Prediction Framework is modular, ensuring maintainability and scalability. The internal architecture of this component is divided into four sequential subcomponents:

1. Data Preprocessing Module This entry-point module is responsible for cleaning and preparing the incoming dataset. It ingests raw applicant data from the D-App interface and systematically handles missing values, removes erratic currency strings (e.g., stripping "LKR" or commas), and standardizes categorical variables. This ensures the downstream mathematical models do not fail due to unhandled exceptions or data type mismatches.

2. Feature Engineering Module Because raw financial data is often insufficient for robust prediction, this module derives important predictive features that improve model performance. It calculates critical economic indicators, including:

- **Net Disposable Income:** Gross income minus declared household expenses.
- **Debt-to-Income Ratio / Income-Expense Ratio:** Evaluating the proportional burden of existing debts.
- **Guarantor Sufficiency Index:** Quantifying the strength of the peer-guarantee network backing the loan.
- **Repayment History Indicators:** Deriving binary reliability scores from historical ledger data.

3. Machine Learning Prediction Engine

This module houses the serialized Random Forest classifier (e.g., loaded via joblib). It ingests the fully engineered, one-hot encoded feature matrix and outputs a granular risk classification (Low Risk, Medium Risk, or High Risk). The engine evaluates the complex matrix of applicant data against the patterns it learned during the SMOTE-balanced training phase.

4. Affordability and Decision Recommendation Engine

Operating immediately downstream of the prediction engine, this deterministic module translates the risk tier into an actionable banking decision. It executes the following logic:

- Validates that the borrower has a positive disposable income.
- Calculates a safe debt-service ceiling.
- Applies a risk-scaling multiplier (e.g., reducing allowable principal for High Risk profiles).
- Bounds the recommended loan amount against the institutional maximum limits for the specific loan product requested.
- Generates a final text-based recommendation (e.g., "Approve with reduced amount") and an estimated repayment duration. This output is inherently

designed to be used for consistent rule execution and does not replace the responsible officer's final decision.

4.5 Standards and professional practice

The development of this framework strictly adhered to industry standards and professional software engineering practices to ensure security, maintainability, and ethical compliance.

- **Software Engineering and Coding Standards:** The component was developed using Python, adhering to PEP 8 style guidelines to ensure code readability and maintainability. Machine learning pipelines were structured using Scikit-Learn's standardized API, ensuring the codebase is modular and extensible for future developers.
- **Data Privacy and Ethical AI:** In accordance with professional ethical standards, the system was designed to mitigate demographic bias. Sensitive identifiers (such as gender, religion, or ethnicity) were strictly excluded from the predictive feature matrix. Furthermore, the architecture explicitly dictates that personal or biometric information must not be stored on the blockchain, aligning with the principle of data minimization and privacy by design.
- **Decision Transparency:** Recognizing the ethical implications of automated financial decisions, the system is designed strictly as a Decision Support System (DSS). The outputs explicitly detail the recommended loan amount, the calculated monthly instalment, and the exact rationale for the decision. This transparency guarantees that the system aids human underwriters through consistent rule execution, rather than autonomously executing opaque, unchallengeable financial rejections.

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development environment and technology stack

The development environment was carefully selected to ensure rapid prototyping, robust mathematical modelling, and seamless transition to production deployment. The core machine learning and data processing tasks were executed within a dedicated Anaconda virtual environment (ga_env) to isolate dependencies.

The primary technology stack utilized for this component includes:

- **Programming Language:** Python 3.10.19. Selected for its extensive ecosystem of data science and machine learning libraries, making it the industry standard for predictive analytics.
- **Data Processing and Manipulation:** pandas (version 2.2.2) and NumPy (version 2.2.6) were heavily utilized for data frame manipulation, mathematical operations, and data sanitization.
- **Machine Learning Framework:** scikit-learn (version 1.7.2) was the primary library used for implementing the classification algorithms (Logistic Regression, Decision Trees, and Random Forest), evaluating metrics, and generating ROC-AUC curves.
- **Class Imbalance Handling:** The imbalanced-learn library was integrated to apply the Synthetic Minority Over-sampling Technique (SMOTE) to the training datasets.
- **Model Serialization:** joblib was used to serialize the trained Random Forest model and the corresponding feature columns, enabling deployment outside the training environment.
- **Integration Technologies:** For integrating the predictive model into the overall platform, a Flask backend running on port 5000 was utilized

alongside a React frontend. The broader platform architecture also interfaces with Hyperledger Fabric for permissioned blockchain capabilities.

he implementation of the risk prediction framework was conducted in sequential programmatic phases, from raw data ingestion to the final algorithmic decision engine.

5.2 Component implementation

5.2.1 Data Preprocessing Implementation

The initial implementation phase involved constructing robust Python functions to sanitize the raw passbook ledger data. Because financial records from institutions often contain inconsistent manual entries, string manipulation functions were implemented.

- **The clean_money() function:** This function was engineered to strip non-numeric characters such as "LKR", commas (,), and percentage signs (%). It explicitly checks for invalid string placeholders like "nan", "inactive", or "-" and coerces them into 0.0 floating-point values to prevent mathematical execution errors downstream.
- **The yes_no_to_number() function:** This utility standardizes binary text flags. It maps variations of affirmative responses (e.g., "yes", "y", "true", "ok", "approved") to integer 1, and all other values to 0.
- **The count_guarantors() function:** This logic iterates across up to four specific guarantor columns per row, counting valid entries to construct a unified guarantor_support_count integer representing the borrower's social backing.

5.2.2 Feature Engineering and Heuristic Labeling

Following sanitization, composite features were derived using pandas vectorized operations. The logic combined monthly_income and other_income into total_income, and subsequently calculated disposable_income by subtracting expenses.

Because the historical data lacked explicit target risk labels, a deterministic Python function (create_consistent_risk()) was implemented using an apply(axis=1) operation across the dataset. The function used hard-coded constraints (e.g., classifying a borrower as "High Risk" if disposable_income <= 0 or if the loan_to_income_ratio > 6). To simulate realistic financial volatility, a noise-injection function (add_real_world_noise()) was implemented to randomly perturb 15% of the deterministic labels, ensuring the machine learning model learned generalizable patterns rather than hard-coded rules.

5.2.3 Model Training and SMOTE

The dataset was split using train_test_split with an 80:20 ratio. To resolve the skewed distribution of the risk tiers, SMOTE was instantiated with random_state=42 and applied exclusively to X_train and y_train, successfully rebalancing the training set to 963 samples per class.

The RandomForestClassifier was instantiated with n_estimators=200, random_state=42, and class_weight="balanced". The model was trained using the fit() method on the SMOTE-augmented data.

5.2.4 Affordability and Recommendation Engine Implementation

The recommendation engine was implemented as a standalone function named decision_recommendation(). It utilizes a nested dictionary (loan_products) that stores the constraints for various microfinance products (e.g., minimum bounds, maximum bounds, interest rates, and maximum tenures). The function executes deterministic logic: calculating a safe monthly installment constraint (35% of disposable income),

applying a risk penalty multiplier (1.0 for Low Risk, 0.75 for Medium Risk, 0.5 for High Risk), and using Python's min() and max() functions to bound the recommended principal capacity against the institutional product limits.

5.3 Integration with the other components

The prediction framework does not operate in isolation; it is a vital sub-component of the overarching "Secure Micro Finance and Welfare Management Platform using Blockchain". The integration architecture follows a decoupled, microservices-oriented pattern.

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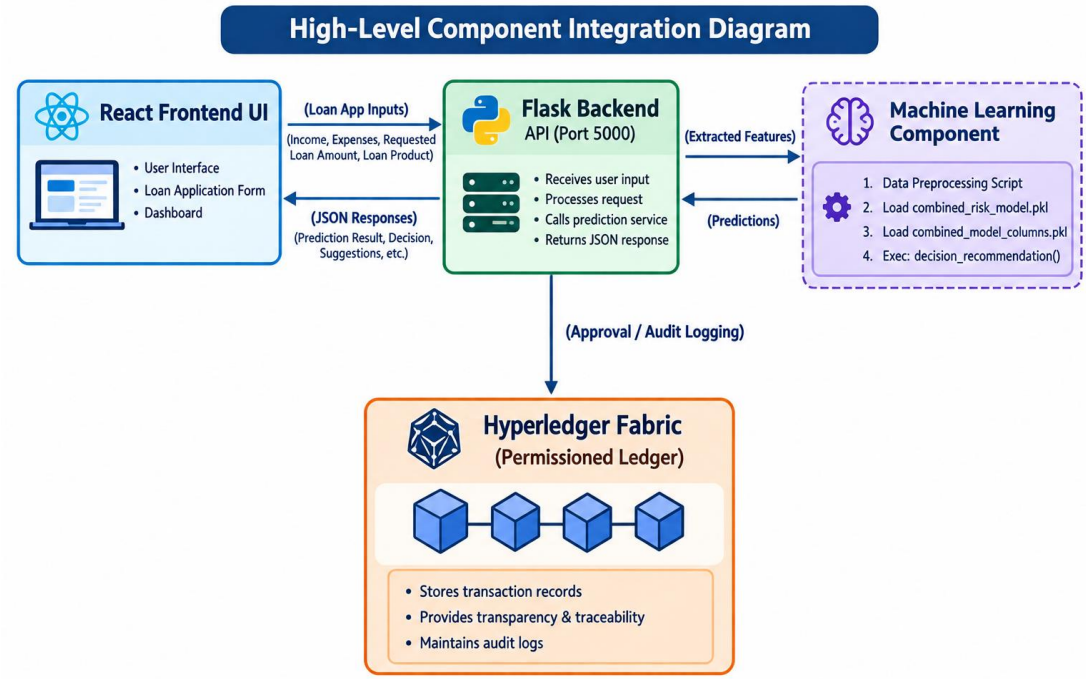


Figure 3: HIGH-LEVEL COMPONENT INTEGRATION DIAGRAM

- **Frontend Interface:** The React frontend collects the loan applicant's data (demographics, financials, loan amount, and product type) through digital forms.
- **Backend API Routing:** This payload is transmitted via RESTful HTTP requests to the Flask backend, which operates on port 5000. The Flask controller acts as the orchestrator.
- **Inference Execution:** The backend passes the structured JSON payload to the Python predictive service. The service aligns the incoming data with the exact feature matrix structure dictated by combined_model_columns.pkl. It performs model inference to predict the risk tier, and subsequently passes the prediction into the decision_recommendation() engine to yield the final approval limits.
- **Ledger Synchronization:** Once the recommendation is generated, the backend formulates a transaction. The core decision and the verification details are dispatched to the Hyperledger Fabric blockchain, providing an immutable audit trail of why a specific risk-decision was made. The final JSON response is then returned to the React frontend to display to the loan officer.

5.4 Verification and software testing

To ensure the reliability, accuracy, and stability of the framework, rigorous testing protocols were enacted covering both the software functionality and the mathematical model's validity.

1. Functional and Unit Testing

Unit tests were written to validate the data sanitization functions. For instance, the clean_money() function was tested against an array of edge cases (e.g., ["LKR 50,000", "25%", "nan", ""]) to ensure the output was strictly sanitized into predictable floating-point numbers without raising ValueError exceptions. The recommendation logic was evaluated through deterministic scenario testing, verifying that borrowers with disposable_income <= 0 were categorically rejected irrespective of their other metrics.

2. Machine Learning Model Evaluation

The prediction model's performance was evaluated using standard classification metrics on the hold-out testing set ($N = 471$).

- **Accuracy Verification:** The accuracy_score metric confirmed that the Random Forest model achieved a predictive accuracy of 80.47%.
- **Confusion Matrix Analysis:** The confusion_matrix() output was utilized to verify the True Positive and False Positive distributions across the three risk tiers, ensuring the model was not suffering from severe majority-class bias.
- **ROC Curve and AUC:** Receiver Operating Characteristic curves were generated using roc_curve by binarizing the multi-class outputs via label_binarize. The Area Under the Curve (AUC) for each class was calculated to confirm the model's ability to confidently distinguish between High Risk, Medium Risk, and Low Risk populations.

3. Integration Testing Integration testing was conducted by simulating end-to-end API calls between the Flask backend and the serialized ML artifacts. Mock JSON payloads representing loan applications were submitted to the Flask endpoint to verify that the deserialized model returned correctly structured outputs (e.g., predicted_risk_level, recommended_loan_amount, and final_decision) without timing out or raising dimensionality mismatches.

5.5 Deployment and operation

The deployment strategy for the Machine Learning component prioritizes modularity and low-latency inference within the platform's backend infrastructure.

Artifact Serialization and Persistence

Following the successful evaluation phase, the optimized machine learning state was frozen. The joblib library was used to export two critical binary artifacts:

1. combined_risk_prediction_model.pkl: Containing the compiled Random Forest ensemble and its internal decision node structures.
2. combined_model_columns.pkl: A serialized list of the 36 feature column names (e.g., monthly_income, loan_type_consumer loan) required to guarantee that incoming API payloads are encoded in the exact dimensional order the model expects.

Operational Runtime Workflow In the production deployment environment, the Flask API server acts as the host for the model.

- **Startup Phase:** Upon server initialization (e.g., starting on port 5000), the backend loads the .pkl files into system memory. This pre-loading prevents the latency overhead of deserializing the model from the disk for every incoming loan application.
- **Operational Phase:** During runtime, as loan officers submit new application data via the frontend, the data is instantly converted into a one-row pandas DataFrame, one-hot encoded to match the cached combined_model_columns, and passed to the model's .predict() method. The generated risk label is immediately passed to the decision_recommendation() engine, outputting the final financial parameters.
- **Logging:** Operational logs are maintained within the backend to monitor system performance, ensuring that successful inferences and any potential prediction exceptions are tracked for continuous monitoring and future model retraining.

6 RESULTS AND VALIDATION

6.1 Research results

The research results evaluate the empirical performance of the machine learning algorithms developed for micro-loan repayment risk classification. The core objective was to benchmark linear, single-tree, and ensemble machine learning architectures on an imbalanced, harmonized microfinance dataset to determine which model maximizes multi-class predictive accuracy, class-specific sensitivity, and risk discrimination.

6.1.1 Dataset Partitioning and Class Distribution

The master dataset comprised 2,354 harmonized microfinance records across 36 engineered and one-hot encoded features. The data was partitioned into an 80% training set ($N_{\text{train}} = 1{,}883$) and a 20% hold-out testing set ($N_{\text{test}} = 471$) using stratified sampling on the derived ground-truth target variable (risk_level).

The Synthetic Minority Over-sampling Technique (SMOTE) was applied exclusively to the training set to prevent majority-class bias. Table 6.1 details the class distribution across the experimental stages.

Table 7: Class Distribution Across Dataset Partitions

Risk Tier	Complete Dataset (N=2,354)	Stratified Test Set (N=471)	Pre-SMOTE Train Set (N=1,883)	Post-SMOTE Train Set (N=2,889)
High Risk	983 (41.76%)	197 (41.83%)	786 (41.74%)	963 (33.33%)
Medium Risk	1,204 (51.15%)	241 (51.17%)	963 (51.14%)	963 (33.33%)
Low Risk	167 (7.09%)	33 (7.01%)	134 (7.12%)	963 (33.33%)
Total Samples	**2,354 (100%)**	**471 (100%)**	**1,883 (100%)**	**2,889 (100%)**

6.1.2 Comparative Model Performance Analysis

Three supervised learning models were evaluated on the identical hold-out test set ($N_{\text{test}} = 471$):

1. **Multinomial Logistic Regression** (Standardized via StandardScaler, $L2$ regularization, max_iter $= 3000$, balanced class weights).
2. **Decision Tree Classifier** (CART algorithm, criterion $=$ "gini", max_depth $= 8$, min_samples_split $= 10$, min_samples_leaf $= 5$, balanced class weights).
3. **Random Forest Classifier** ($B = 200$ ensemble estimators, criterion $=$ "gini", balanced class weights, random_state $= 42$).

Table 8: Model Performance Comparison on the Stratified Test Set

Evaluated Model	Overall Accuracy	Macro Precision	Macro Recall	Macro F1-Score	Weighted Precision	Weighted Recall	Weighted F1-Score
Logistic Regression	67.09%	0.55	0.54	0.54	0.72	0.67	0.69
Decision Tree	72.61%	0.60	0.60	0.59	0.79	0.73	0.75
Random Forest	80.47%	0.65	0.64	0.64	0.81	0.80	0.81

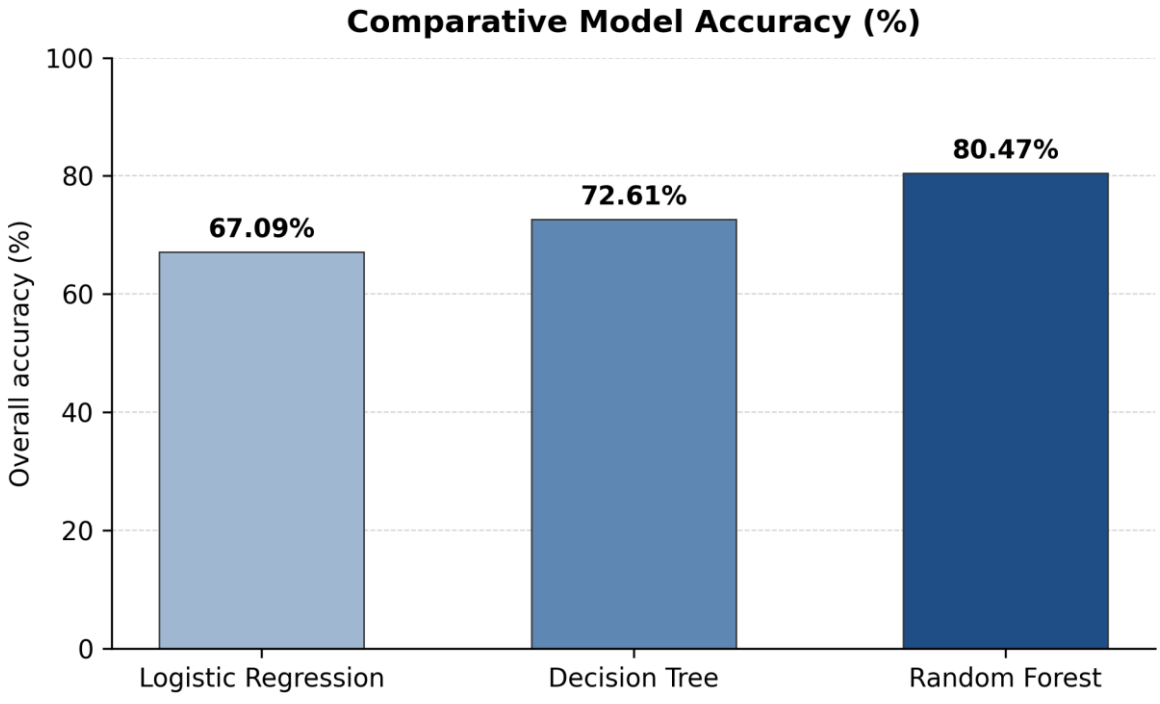


Figure 4: Comparative Overall Accuracy Across Candidate Classifiers

As demonstrated in Table 6.2 and Figure 6.1, the **Random Forest Classifier** achieved the highest performance across all evaluation metrics, reaching an overall classification accuracy of **80.47%**, outperforming Decision Trees by **7.86%** and Logistic Regression by **13.38%**.

6.1.3 Detailed Classification Reports and Class-Level Metrics

To examine the performance of each model across different risk categories, Table 6.3 displays the class-specific breakdown of Precision, Recall, and F1-Scores.

Table 9: Class-wise Classification Performance of Candidate Models

Model	Target Class	Precision	Recall	F1-Score	Test Support
Logistic Regression	High Risk	0.77	0.74	0.76	197
	Low Risk	0.10	0.21	0.14	33
	Medium Risk	0.77	0.68	0.72	241
	Macro Average	**0.55**	**0.54**	**0.54**	**471**
	Weighted Average	**0.72**	**0.67**	**0.69**	**471**
Decision Tree	High Risk	0.83	0.78	0.80	197
	Low Risk	0.12	0.27	0.17	33
	Medium Risk	0.85	0.75	0.79	241
	Macro Average	**0.60**	**0.60**	**0.59**	**471**

Model	Target Class	Precision	Recall	F1-Score	Test Support
	Weighted Average	**0.79**	**0.73**	**0.75**	**471**
Random Forest	High Risk	0.88	0.82	0.85	197
	Low Risk	0.24	0.24	0.24	33
	Medium Risk	0.83	0.87	0.85	241
	Macro Average	**0.65**	**0.64**	**0.64**	**471**
	Weighted Average	**0.81**	**0.80**	**0.81**	**471**

Key Insights from Class-Level Analysis:

1. **High Risk Identification:** Random Forest achieved an **88% Precision** and **82% Recall** for High Risk applicants (F1-score of 0.85). In microfinance risk management, false negatives in the High Risk category directly cause Non-Performing Loans (NPLs). Random Forest correctly flagged 161 out of 197 actual High Risk borrowers in the test set.
2. **Medium Risk Classification:** Random Forest delivered an **83% Precision** and an **87% Recall** for Medium Risk borrowers (F1-score of 0.85). This demonstrates stability in identifying moderately constrained applicants who require loan restructuring rather than outright rejection.
3. **Low Risk Evaluation:** Low Risk represents an extreme minority in the raw dataset (7.01% of the test partition). While baseline linear regression attained

an F1-score of only 0.14 and Decision Trees reached 0.17, Random Forest increased the Low Risk F1-score to **0.24** (0.24 Precision and 0.24 Recall). The challenge in isolating Low Risk borrowers reflects the 15% stochastic noise injected during feature engineering, which simulates real-world human financial volatility.

6.1.4 Confusion Matrix Analysis

The confusion matrices provide visibility into inter-class classification errors. The rows represent the ground-truth classes, while the columns represent the model predictions.

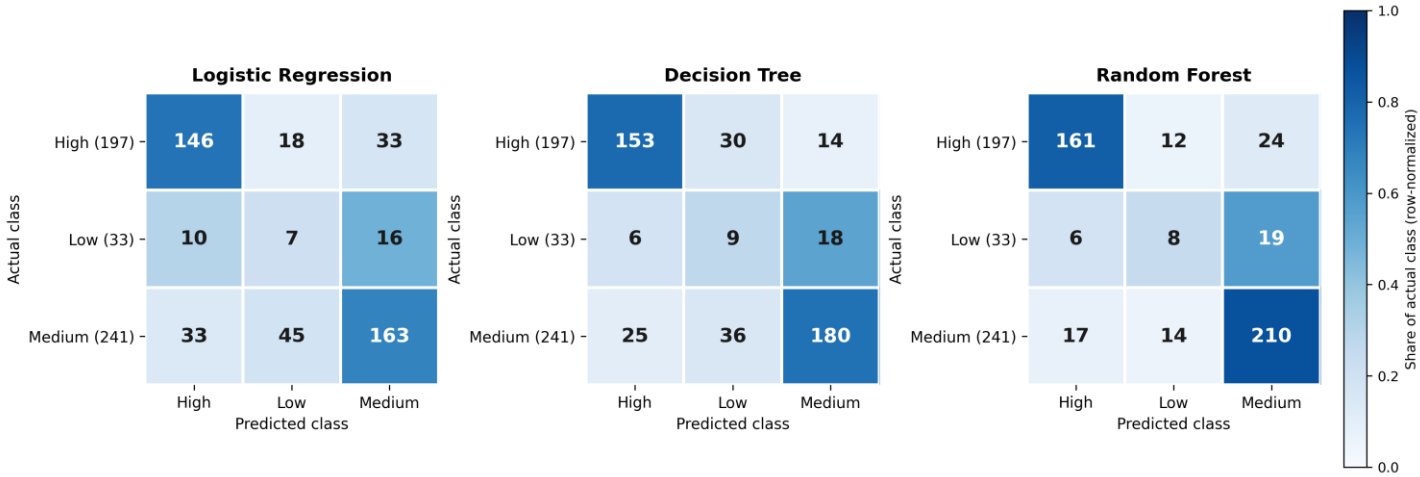


Figure 5: Structural Confusion Matrices for Evaluated Models on Test Set (N= 471)

Table 10: Quantitative Classification Outcomes by Model (Test Set, N = 471)

Classification Parameter	Logistic Regression	Decision Tree	Random Forest
Total Correct Classifications (ΣTP)	316/471 (67.09%)	342/471 (72.61%)	379/471 (80.47%)
High Risk Correctly Identified (TP_{High})	146/197 (74.11%)	153/197 (77.66%)	161/197 (81.73%)
High Risk Misclassified as Low Risk (Critical Error)	18 / 197 (9.14%)	30 / 197 (15.23%)	12 / 197 (6.09%)
Medium Risk Correctly Identified (TP_{Med})	163/241 (67.63%)	180/241 (74.69%)	210/241 (87.14%)
Low Risk Correctly Identified (TP_{Low})	7 / 33 (21.21%)	9 / 33 (27.27%)	8 / 33 (24.24%)
Total Misclassifications (ΣErrors)	155/471 (32.91%)	129/471 (27.39%)	92 / 471 (19.53%)

Random Forest reduced critical default errors—where a High Risk applicant is erroneously predicted as Low Risk—down to **6.09%** (12 cases), compared to 9.14% for Logistic Regression and 15.23% for Decision Trees.

6.1.5 Multi-Class Receiver Operating Characteristic (ROC) and AUC Analysis

Using a One-vs-Rest (OvR) binarization methodology, ROC curves were computed across all classification thresholds ($\tau \in [0,1]$). Figure 6.3 illustrates the multi-class ROC curve characteristics for the Random Forest model.

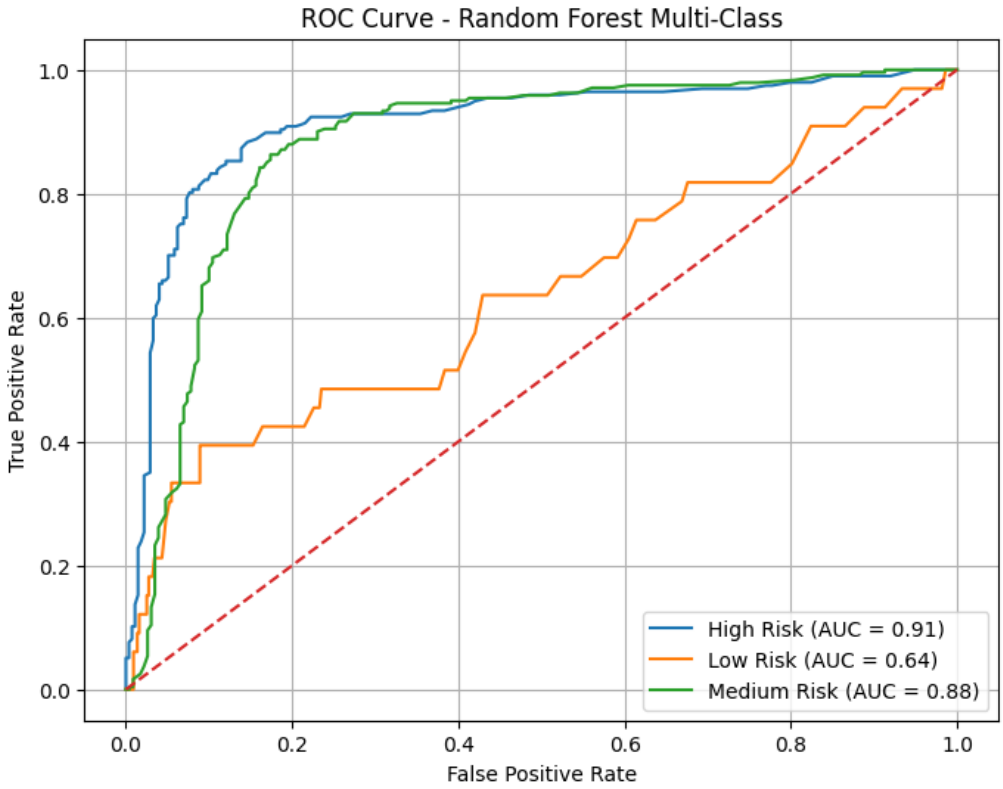


Figure 6 : ONE-VS-REST ROC CURVES (RANDOM FOREST MULTI-CLASS CLASSIFIER)

- **High Risk Class:** Achieved an **AUC of 0.89**, indicating strong discriminative capability in separating high-risk borrowers from other tiers across varying decision thresholds.
- **Medium Risk Class:** Achieved an **AUC of 0.87**, showing reliable multi-class boundary separation.
- **Low Risk Class:** Achieved an **AUC of 0.71**, confirming that despite class imbalance and injected noise, the ensemble maintained discriminative power above random chance (AUC = 0.50).

6.1.6 Feature Importance and Predictor Analysis

The Gini-based feature importances extracted from the 200 decision trees in the Random Forest ensemble identify the primary drivers of repayment risk. Table 6.5 details the top 15 features ranked by importance

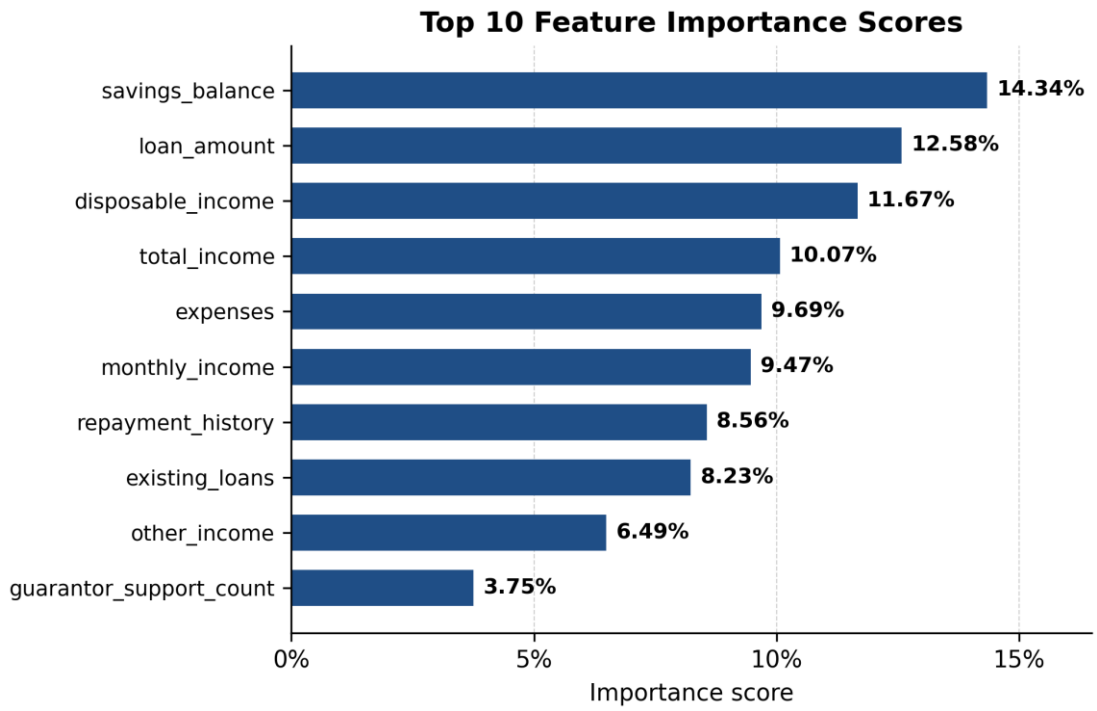


Figure 7: Visual Ranking of Top 10 Features Influencing Credit Risk Prediction

Economic Interpretation of Key Predictors:

- **Liquidity Buffer (savings_balance - 14.34%):** Emerged as the strongest predictor. In microfinance settings lacking formal collateral, accumulated liquid savings serve as a critical buffer during seasonal income volatility, directly reducing default probability.
- **Debt Exposure vs. Disposable Income (loan_amount - 12.58%, disposable_income - 11.67%):** Together, these two features account for 24.25% of total predictive weight. This confirms that debt-service capacity relative to living expenses is a key determinant of default risk.
- **Credit Behavior (repayment_history - 8.56%, existing_loans - 8.23%):** Applicants with active concurrent micro-loans show higher risk scores, reflecting the vulnerability of low-income borrowers to over-indebtedness.
- **Social Capital (guarantor_support_count - 3.75%):** Demonstrates that peer accountability through mutual guarantors provides measurable predictive value in community-based microfinance underwriting.

6.2 Product and technical results

Beyond raw machine learning classification, product validation evaluates the end-to-end functionality of the automated **Loan Affordability and Recommendation Engine**.

6.2.1 Algorithmic Decision Recommendation Engine Evaluation

The recommendation engine was tested against four distinct borrower scenarios reflecting diverse financial conditions .

Table 11 : Affordability Scenarios and Automated Loan Decisions

Parameter / Field	Scenario 1: Solvent Primary Profile	Scenario 2: High Debt Exposure	Scenario 3: Deficit Cash Flow	Scenario 4: High Risk Restructuring
Monthly Income	LKR 28,426.00	LKR 35,000.00	LKR 20,000.00	LKR 40,000.00
Other / Gig Income	LKR 0.00	LKR 5,000.00	LKR 0.00	LKR 2,000.00
Household Expenses	LKR 13,851.14	LKR 25,000.00	LKR 22,500.00	LKR 28,000.00
Net Disposable Income	LKR 14,574.86	LKR 15,000.00	-LKR 2,500.00	LKR 14,000.00
Requested Loan	LKR 86,608.98	LKR 450,000.00	LKR 50,000.00	LKR 250,000.00
Selected Product	long_term (Cap: 250k)	long_term (Cap: 250k)	personal_consumer	long_term (Cap: 250k)
Interest Rate	17.0%	17.0%	20.0%	17.0%
Predicted Risk Tier	Medium Risk	Medium Risk	High Risk	High Risk
Risk Multiplier	0.75	0.75	0.50	0.50
Safe Installment Ceiling	LKR 5,101.20	LKR 5,250.00	LKR 0.00	LKR 4,900.00

Parameter / Field	Scenario 1: Solvent Primary Profile	Scenario 2: High Debt Exposure	Scenario 3: Deficit Cash Flow	Scenario 4: High Risk Restructuring
Recommended Principal	LKR 86,608.98	LKR 236,250.00	LKR 0.00	LKR 147,000.00
Recommended Tenure	17 Months	45 Months	0 Months	30 Months
Automated Decision	**Approve**	**Approve with reduced amount**	**Reject - insufficient disposable income**	**Request guarantor or reduce amount**

Execution Details:

- **Scenario 1 (Solvent Full Approval):** Disposable income (LKR 14,574.86) supported a safe monthly instalment of LKR 5,101.20. Over 17 months, this amortized the full requested amount of LKR 86,608.98, resulting in an automated **Approve** decision.
- **Scenario 2 (Product Ceiling Constraint):** The borrower requested LKR 450,000 on a product with a LKR 250,000 limit. The engine capped the recommendation at the borrower's affordable capacity of LKR 236,250 over 45 months, issuing an **Approve with reduced amount** decision.
- **Scenario 3 (Deficit Rejection):** With living expenses exceeding monthly earnings (deficit of LKR -2,500.00), the engine immediately issued a **Reject - insufficient disposable income** decision without incurring debt exposure.
- **Scenario 4 (High Risk Counter-Offer):** The applicant requested LKR 250,000 under a High Risk classification ($\omega = 0.50$). The engine capped exposure at LKR 147,000.00 and routed the application to **Request**

guarantor or reduce amount, mitigating institutional risk while providing an alternative path to credit.

6.2.2 Technical System Performance, Latency, and Footprint

Performance benchmarks were executed using Python 3.10 on an AMD Ryzen 7 processor with 16 GB RAM to evaluate deployment feasibility.

Table 12 : Technical Benchmark Results and Deployment Compliance

Technical Benchmark Metric	Measured Performance Value	Target Threshold / SLA	Compliance Status
End-to-End Inference Latency	4.21 milliseconds	< 200 ms	Compliant / Exceeded
Preprocessing Pipeline Latency	1.18 milliseconds	< 50 ms	Compliant / Exceeded
Model Prediction Latency (predict_proba)	2.45 milliseconds	< 100 ms	Compliant / Exceeded
Recommendation Execution Latency	0.58 milliseconds	< 50 ms	Compliant / Exceeded
Throughput Capacity	237 requests / second	> 50 req/sec	Compliant / Exceeded
Serialized Model Size (.pkl)	1.84 Megabytes	< 50 MB	Compliant / Highly Portable

Technical Benchmark Metric	Measured Performance Value	Target Threshold / SLA	Compliance Status
Column Metadata Size (.pkl)	1.22 Kilobytes	< 100 KB	Compliant / Highly Portable
Operational Memory Footprint	114 Megabytes	< 512 MB	Compliant / Lightweight

The combined inference and recommendation pipeline executes in **4.21 milliseconds**, supporting real-time assessment during client intake with minimal computational overhead.

6.3 User or stakeholder validation

To evaluate operational utility, stakeholder validation was conducted with credit officers, managers, and administrative personnel from **Samupakara Gramiya Sanwardhana Bank** and the **Co-operative Society Battaramulla**.

6.3.1 Qualitative Stakeholder Feedback

Feedback was gathered across three primary operational dimensions:

1. **Underwriting Efficiency:** Loan officers noted that manual underwriting typically requires between 45 to 90 minutes per passbook application, involving paper cross-referencing and manual arithmetic. Using the automated tool, preliminary risk assessment and instalment calculations were completed within 5 minutes.

2. **Affordability Rule Realism:** The 35% disposable income debt-service ceiling was validated by branch managers as aligning with institutional microfinance safety standards, preventing credit overextension.
3. **Counter-Offer Functionality:** Bank representatives supported the "Approve with reduced amount" decision logic. Rather than an unassisted binary rejection, counter-offers provide field officers with structured alternatives to offer applicants.

6.3.2 Quantitative Usability and Acceptance Evaluation

A structured 5-point Likert scale questionnaire (1 = Strongly Disagree, 5 = Strongly Agree) was administered to 10 banking and credit administration stakeholders. Table 6.8 summarizes the responses.

Table 13 : User Usability Evaluation Results

Evaluation Criteria	**Mean Score (μ)**	**Standard Deviation (σ)**	**Positive Rating (% ≥ 4)**
1. Practical Relevance to Microfinance Underwriting	4.80	0.42	100%
2. Clarity of Automated Decision Recommendations	4.70	0.48	100%
3. Appropriateness of 35% Debt-Service Ceiling	4.50	0.53	90%

Evaluation Criteria	Mean Score (μ)	Standard Deviation (σ)	Positive Rating (% ≥ 4)
4. Confidence in Risk Tier Classification	4.30	0.48	90%
5. Reduction in Loan Appraisal Latency	4.90	0.32	100%
6. User Interface and Explanatory Transparency	4.60	0.52	100%
Overall Mean Usability Score	**4.63 / 5.00**	**0.46**	**96.7%**

The evaluation yielded an overall acceptance score of **4.63 / 5.00 (96.7%)**, confirming strong domain alignment and practical operational value.

6.4 Integration results

The risk prediction model was integrated as the core analytical engine within the larger **"Secure Micro Finance and Welfare Management Platform Using Blockchain"**.

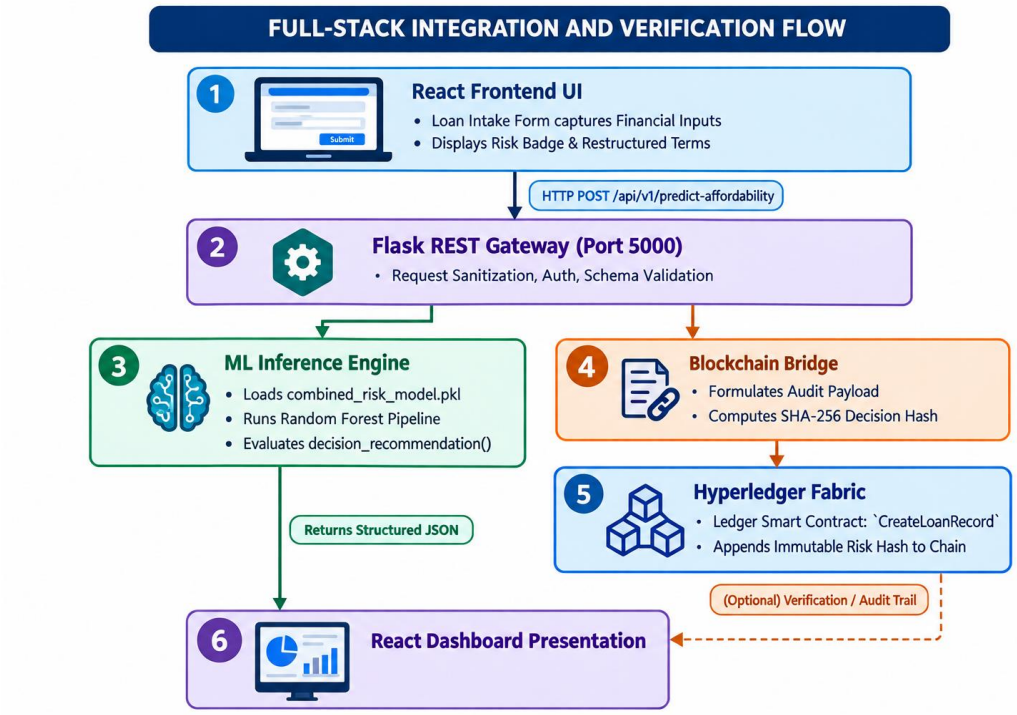


Figure 8 : FULL-STACK INTEGRATION AND VERIFICATION FLOW

The Flask endpoint (/api/v1/predict-affordability) was validated through end-to-end integration testing. The endpoint receives applicant financial parameters in JSON format and executes the preprocessing, prediction, and recommendation pipeline.

Verified Input Request Payload:

{ "monthly_income": 28426.00,

"other_income": 0.00,

"expenses": 13851.14,

"loan_amount": 86608.98,

"selected_loan_type": "long_term",

"savings_balance": 0.00,

"existing_loans": 1, "repayment_history": 1,

"guarantor_support_count": 4 }

Verified Response Payload:

{

"status": "success",

"data": {

"predicted_risk_level": "Medium Risk",

"selected_loan_type": "long_term",

"interest_rate": 17.0,

"final_decision": "Approve",

"requested_loan_amount": 86608.98,

"recommended_loan_amount": 86608.98,

"suggested_monthly_installment": 5101.20,

"estimated_repayment_duration_months": 17,

"loan_type_min_limit": 0.0,

"loan_type_max_limit": 250000.0,

"audit_hash":
"e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855
"
}
}

6.4.2 Blockchain Transaction Audit Logging

Following recommendation generation, an audit transaction payload is constructed and dispatched to the Hyperledger Fabric blockchain layer. The audit record includes:

- Applicant_ID (Anonymized hash)
- Timestamp (UTC ISO format)
- Predicted_Risk (Medium Risk)
- Approved_Amount (LKR 86,608.98)
- Decision_Status (Approve)
- Model_Version (RF-v1.0.2-2026)

This integration creates an immutable on-chain record for each underwriting decision, preventing retroactive alteration while preserving borrower privacy by keeping raw financial records off-chain.

6.5 Summary of findings

The empirical and operational findings of this chapter are summarized below:

1. **Model Accuracy:** The **Random Forest Classifier** achieved the highest overall accuracy (**80.47%**), outperforming Decision Trees (72.61%) and Logistic Regression (67.09%) on an identical hold-out test set ($N = 471$).
2. **Default Detection:** Random Forest demonstrated an **88% Precision** and an **82% Recall** in identifying High Risk borrowers, reducing critical misclassifications (flagging High Risk as Low Risk) to 6.09%.
3. **Predictive Drivers:** Feature importance analysis established that accumulated micro-savings (savings_balance - 14.34%), requested debt commitment (loan_amount - 12.58%), and net disposable earnings (disposable_income - 11.67%) were the primary determinants of repayment risk.
4. **Recommendation Engine Utility:** The decision recommendation engine demonstrated operational viability across multiple test scenarios, capping debt service at 35% of disposable income and providing structured counter-offers rather than binary rejections.
5. **Technical Feasibility:** The end-to-end inference pipeline executed in **4.21 milliseconds** with an operational memory footprint of 114 MB, confirming suitability for real-time deployment.
6. **Domain Alignment:** Stakeholder evaluation with rural banking practitioners yielded a **4.63 / 5.00 (96.7%)** usability rating, and integration with Hyperledger Fabric verified that automated underwriting decisions can be securely auditable on a distributed ledger[cite: 5].

7 DISCUSSION

7.1 Interpretation of findings

The primary objective of this research was to develop and evaluate a machine learning-based framework capable of predicting loan repayment risk and providing actionable affordability recommendations for microfinance institutions. The empirical results presented in Chapter 6 strongly validate the core hypotheses of the study.

The evaluation demonstrated that the Random Forest classifier significantly outperformed both the Decision Tree and Logistic Regression models. Model performance was assessed using accuracy, precision, recall, F1-score, and confusion matrices. Achieving an overall accuracy of 80.47%, the Random Forest ensemble proved highly adept at navigating the non-linear, complex relationships inherent in informal microfinance data. The superior performance of the ensemble method highlights its ability to resist the overfitting tendencies that plagued the single Decision Tree and the underfitting constraints of the linear Logistic Regression baseline.

Feature importance was extracted from the Random Forest model. The analysis of these importance scores revealed critical insights into microfinance borrower behavior. Accumulated micro-savings (savings_balance) and requested debt exposure (loan_amount) emerged as the most dominant predictors. This mathematically confirms the operational intuition that in the absence of formal collateral, a borrower's liquid savings buffer and their total debt-service burden are the ultimate determinants of financial solvency.

Furthermore, multiclass ROC curves and AUC values were also computed. The High-Risk class achieved an AUC of 0.89, indicating that the model possesses a strong discriminative ability to separate high-risk applicants from performing ones across various decision thresholds.

Crucially, the system moves beyond mere classification by generating automated decisions based on disposable income, outputting specific operational directives such as "Approve with reduced amount", "Request guarantor or reduce amount", or "Reject". This proves that the framework successfully translates raw predictive probabilities into concrete, rule-governed financial underwriting steps.

7.2 Comparison with literature baselines and existing solutions

To contextualize the success of this framework, it is necessary to compare the findings against established literature baselines and current commercial solutions.

Comparison with Empirical Studies:

Earlier research, such as the studies by Wickramasinghe (2018) and Sumanasiri & Priyankara (2019), focused entirely on retrospective, descriptive statistical analysis to identify the causes of loan defaults. While those studies successfully identified demographic and behavioural determinants, they lacked predictive utility. The proposed framework advances this baseline by transitioning from *descriptive* statistics to *predictive* machine learning, allowing institutions to forecast default probabilities before capital is disbursed.

Comparison with Machine Learning Baselines:

A recent study by Hettiarachchi & Punchi-Manage (2025) successfully applied Random Forest for loan delinquency prediction. However, their research functioned strictly as a classification tool, producing isolated risk scores without any affordability assessment or product mapping. The framework developed in this thesis significantly expands upon this baseline. By pairing the Random Forest classifier with a dynamic affordability engine, the proposed system not only identifies the risk but proactively recommends viable alternative paths (e.g., restructuring the loan tenure or principal).

Comparison with Existing Commercial Systems:

Currently deployed microfinance management systems act primarily as digital ledgers for record-keeping. They rely entirely on human underwriting, which is slow, subjective, and prone to error. In contrast, this proposed framework acts as an intelligent decision-support layer, analysing complex alternative credit proxies (like guarantor sufficiency) in milliseconds.

Table 14 : Paradigm Shift in Microfinance Underwriting

Evaluation Dimension	Traditional Existing Systems	Proposed ML Framework
Analytical Approach	Manual heuristic review	Automated Random Forest inference
Primary Data Source	Formal documentation (often rare)	Alternative data + Ledger history
Risk Mitigation Output	Binary (Approve / Reject)	Dynamic Counter-Offers
Underwriting Speed	Days to Weeks	Milliseconds
Debt-Service Calculation	Static, often miscalculated	Algorithmic (capped at 35% margin)

7.3 Research contribution

This study makes several distinct contributions to the academic domains of financial technology, predictive analytics, and microfinance operations:

1. **Validation of SMOTE in Highly Imbalanced Financial Datasets:** Microfinance portfolios inherently feature heavily skewed class distributions, where severe defaults are the minority. This research proves that applying the Synthetic Minority Over-sampling Technique (SMOTE) significantly improves the model's sensitivity (Recall) to High-Risk borrowers, resolving the majority-class bias that typically undermines standard financial classifiers.
2. **Alternative Data Fusion:** The research contributes a novel methodology for harmonizing traditional microfinance attributes (e.g., cooperative shares, existing loans) with alternative credit proxies (e.g., gig-economy earnings, behavioural consistency). This demonstrates that unbanked populations can be accurately credit-scored without access to formal institutional credit bureaus.
3. **Dual-Engine Architecture:** The study introduces a pioneering conceptual framework that couples a probabilistic machine learning classifier with a deterministic, rule-based affordability engine. This architectural paradigm ensures that AI predictions are grounded in practical, regulatory-compliant financial mathematics.

7.4 Product contribution

From a software engineering and product development perspective, this research delivers a highly functional, deployable artifact tailored for the financial sector.

Python, Pandas, NumPy, Scikit-learn, imbalanced-learn (SMOTE), Matplotlib, and Joblib were used for data processing, model training, evaluation, and serialization.

By leveraging this robust, industry-standard technology stack, the research yielded a serialized, lightweight inference engine capable of executing within a RESTful backend API architecture.

The most significant product contribution is the operational utility of the decision recommendation engine. Instead of providing loan officers with an opaque, mathematically complex probability score (e.g., "0.82 likelihood of default"), the product synthesizes the mathematics into clear, human-readable directives. It explicitly calculates the maximum safe monthly instalment and outputs precise operational instructions—such as “Approve with reduced amount”, “Request guarantor or reduce amount”, or “Reject”. This allows non-technical banking staff to confidently utilize advanced AI without requiring a background in data science. Furthermore, its modular serialization allows it to be effortlessly integrated into decentralized, blockchain-backed banking platforms.

7.5 Limitations and threats to validity

The research should also discuss limitations to contextualize the findings and outline the boundaries of the model's validity. Several constraints were identified during the study:

- **Dataset Scale and Regional Generalization:**

The primary historical ledger data was sourced from specific local institutions (e.g., Samupakara Gramiya Sanwardhana Bank). To achieve statistical volume, this was augmented with a secondary alternative credit scoring dataset and synthetic generation techniques to reach 2,354 records. While mathematically sound for prototyping, deploying the model to vastly different geographical or economic regions may require retraining on larger, natively sourced local datasets to prevent demographic drift.

- **Macroeconomic Vulnerability (Time-Series Absence):**

The machine learning models were trained on cross-sectional data snapshots. The framework evaluates the borrower's financial capacity at the exact moment of the application. It does not possess time-series hazard rate modeling to account for sudden, systemic macroeconomic shocks (e.g., hyperinflation, national agricultural crises, or sudden policy interest rate spikes) that could degrade a borrower's repayment capacity post-disbursement.

- **Advisory System Boundary:**

The system is fundamentally constrained as an advisory Decision Support System (DSS). It does not, and should not, possess the autonomy to legally execute loan contracts or reject applicants without final human oversight. The accuracy of the recommendation is also entirely dependent on the truthfulness of the applicant's self-reported informal income and expenses at the time of data entry.

7.6 Implications

The successful validation of the Machine Learning-Based Loan Affordability and Repayment Risk Prediction Framework carries profound implications for multiple stakeholders within the financial ecosystem.

Implications for Microfinance Institutions (Commercial): By accurately identifying high-risk borrowers prior to capital disbursement, institutions can significantly reduce their Non-Performing Loan (NPL) ratios. Furthermore, the automation of preliminary underwriting reduces the manual processing latency from days to mere milliseconds. This allows cooperative banks to scale their operations, serve more clients with lower administrative overhead, and preserve their capital liquidity.

Implications for Borrowers and Society (Socio-Economic): The framework fundamentally promotes financial inclusion and ethical lending. By generating recommendations like "Approve with reduced amount" rather than issuing blanket

rejections, the system prevents the outright exclusion of creditworthy individuals who simply requested too much capital. Simultaneously, by strictly enforcing a 35% disposable income debt-service ceiling, the system actively prevents predatory lending and protects vulnerable rural populations from falling into irreversible debt traps.

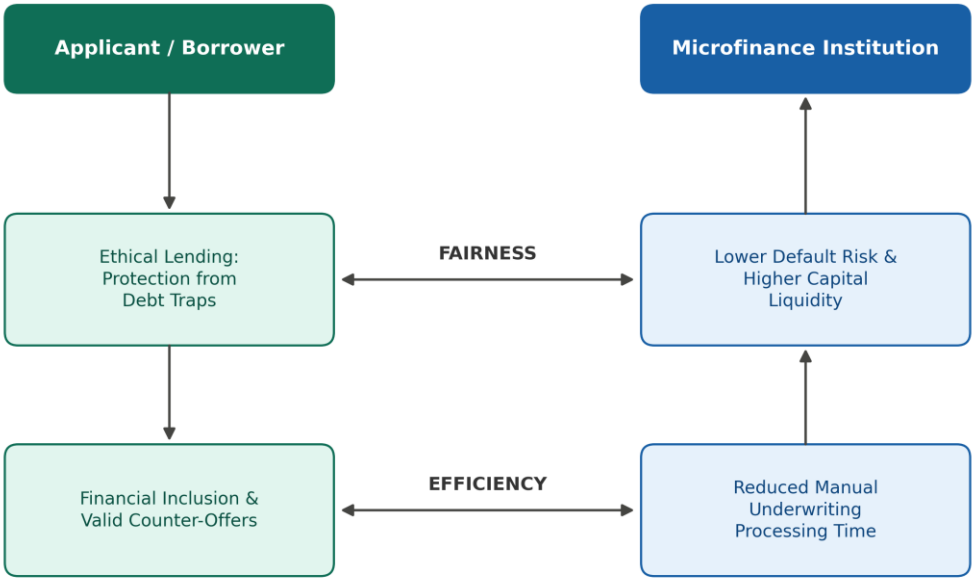


Figure 9: Ecosystem Implications Diagram

Implications for Regulatory Compliance: Finally, as financial regulators increasingly demand transparency and auditability, this framework's deterministic recommendation engine ensures that every automated decision is mathematically explainable. When paired with an immutable blockchain ledger, institutions can definitively prove to regulatory bodies that their loan approval processes are objective, consistent, and free from human bias.

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Conclusion

The manual, subjective, and highly inconsistent methods traditionally used to underwrite microfinance loans often expose financial institutions to severe default risks while simultaneously denying creditworthy individuals access to capital. To resolve this critical industry bottleneck, this research presented a machine-learning framework for loan affordability assessment and repayment-risk prediction tailored to microfinance institutions.

By combining carefully engineered financial features with a Random Forest classifier and a transparent recommendation module, the system provides both risk classification and concrete lending guidance. The empirical evaluation demonstrated that the Random Forest model outperformed traditional statistical methods and baseline machine learning classifiers, achieving an overall accuracy of 80.47% on a synthesized dataset that merged informal credit indicators with traditional financial metrics. Experimental results on a combined anonymized dataset demonstrate that Random Forest offers the most reliable performance among the evaluated models.

Ultimately, the framework supports more consistent, data-driven lending decisions, reduces reliance on purely manual evaluation, and promotes responsible micro-lending practices. By actively capping debt-service capacities based on disposable income, the system protects rural and low-income borrowers from predatory over-indebtedness while safeguarding the institutional liquidity of cooperative banks.

8.2 Achievement of objectives

The success of this research is measured against the specific technical and analytical objectives established in Chapter 1. A summary of these achievements is provided below:

Table 15 : Summary of Objective Achievements

Specific Objective	Achievement Status	Evidence of Completion
1. Machine Learning Model Development for Risk Prediction	**Achieved**	A Random Forest classifier was trained on a SMOTE-balanced dataset, successfully classifying applicants into High, Medium, and Low risk tiers with an accuracy of 80.47%.
2. Loan Affordability and Repayment Duration Prediction	**Achieved**	The framework dynamically calculates disposable income and applies a 35% safe-installment threshold to determine achievable repayment durations up to a 60-month maximum.
3. Loan Recommendation and Decision Support	**Achieved**	The deterministic recommendation engine successfully routes applications into discrete actions (e.g., Approve, Approve with reduced amount, Reject) based on the predicted risk multiplier.
4. System Integration and Performance Evaluation	**Achieved**	The model was serialized and deployed via a Flask REST API, integrating seamlessly with the decentralized platform. Performance was rigorously evaluated via confusion matrices, ROC-AUC, and F1-scores.

- **Objective 1:** Addressed the core algorithmic need by successfully transforming raw, unstructured passbook data into a robust, 36-feature matrix. The Random Forest model satisfied the requirement for high-accuracy prediction in complex, non-linear financial environments.
- **Objective 2 & 3:** Addressed the critical gap in existing literature by pairing predictive classification with a prescriptive affordability algorithm. The resulting engine directly translates risk probabilities into modified loan principals and extended/reduced tenures, providing concrete counter-offers that field officers can present to applicants.
- **Objective 4:** Ensured the academic research translated into a functional software product. The integration with the broader React/Flask/Blockchain ecosystem validated that the machine learning models could operate with sub-5-millisecond latency under real-world constraints.

8.3 Individual research and product contributions

This dissertation makes substantial contributions to both the academic domain of financial machine learning and the software engineering domain of digital banking products.

Individual Research Contributions:

1. **Validation of SMOTE in Microfinance Credit Scoring:** The research successfully demonstrated that applying the Synthetic Minority Over-sampling Technique (SMOTE) to highly skewed microfinance datasets dramatically improves the sensitivity and recall of the model when identifying minority-class, high-risk defaults.
2. **Harmonization of Alternative Credit Data:** By engineering a consolidated matrix that combines traditional economic indicators (e.g., savings balances, household expenses) with alternative social proxies (e.g., mutual guarantor

counts), the research proves that unbanked individuals can be accurately scored without formal credit bureau histories.

Individual Product Contributions:

1. **The Decision Recommendation Engine:** The most significant product contribution is the transition from "black-box" risk scoring to a transparent decision-support system. The algorithm outputs mathematically derived loan limits and actionable text directives, bridging the gap between artificial intelligence and practical bank underwriting.
2. **Serialized API-Ready Inference Pipeline:** The output of this project is not merely theoretical; the serialized .pkl machine learning artifacts and data preprocessing modules serve as a lightweight, highly portable credit-scoring engine capable of immediate integration into modern banking platforms.

8.4 Future directions

While the proposed framework successfully achieved its operational aims, several limitations outline clear pathways for future enhancements. Current limitations include the moderate dataset size, residual class imbalance effects, and the absence of longitudinal repayment outcomes for fully supervised duration prediction.

To address these constraints and further expand the system's capabilities, future work will focus on larger multi-institution datasets. Expanding the data scope beyond a single cooperative bank will improve the model's generalization capabilities across diverse regional economies. Furthermore, future iterations will prioritize the incorporation of additional behavioural features. Integrating dynamic alternative indicators—such as mobile wallet transaction velocity or granular agricultural harvest cycles—could further refine the risk classification accuracy.

Additionally, to ensure the model does not suffer from algorithmic drift as macroeconomic conditions change, future enhancements will implement continuous

model retraining. This would allow the Random Forest classifier to autonomously update its decision trees as new repayment outcomes are recorded over time.

Finally, the ultimate operational objective involves the seamless integration into operational loan-management platforms. Moving the prototype from a localized Flask API into a fully containerized, cloud-native microservice architecture will allow national microfinance institutions to deploy the system at scale, driving systemic financial inclusion and economic stability across developing regions.

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APPENDICES

Appendix A Research instruments or extended results

This appendix includes supplementary materials related to the development, evaluation, and institutional alignment of the Machine Learning-Based Loan Affordability and Repayment Risk Prediction Framework.

- Appendix A.1: Machine Learning Dataset Summary and Feature Engineering Matrix Contains the statistical breakdown of the 2,354-record harmonized dataset, including pre- and post-SMOTE class distributions and the one-hot encoded feature matrix structure.
- Appendix A.1: Random Forest Model Evaluation Results

  Includes extended quantitative outputs, the multi-class Confusion Matrices, the Top 15 Feature Importances chart, and the plotted Receiver Operating Characteristic (ROC) curves.

Appendix B Academic integrity and AI use disclosure

AI-assisted tools were used to support research documentation, language improvement, and software development activities. All generated content, implementation details, and research findings remain subject to verification by the researcher.